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(54) **SHARED DECODER ARCHITECTURE FOR THREE-DIMENSIONAL MEMORY ARRAYS**

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(71) Applicant: **Micron Technology, Inc.**, Boise, ID (US)

(72) Inventors: **Christophe Vincent Antoine Laurent**, Agrate Brianza (IT); **Andrea Martinelli**, Bergamo (IT); **Efrem Bolandrina**, Fiorano al Serio (IT); **Ferdinando Bedeschi**, Biassono (IT)

(57) **ABSTRACT**

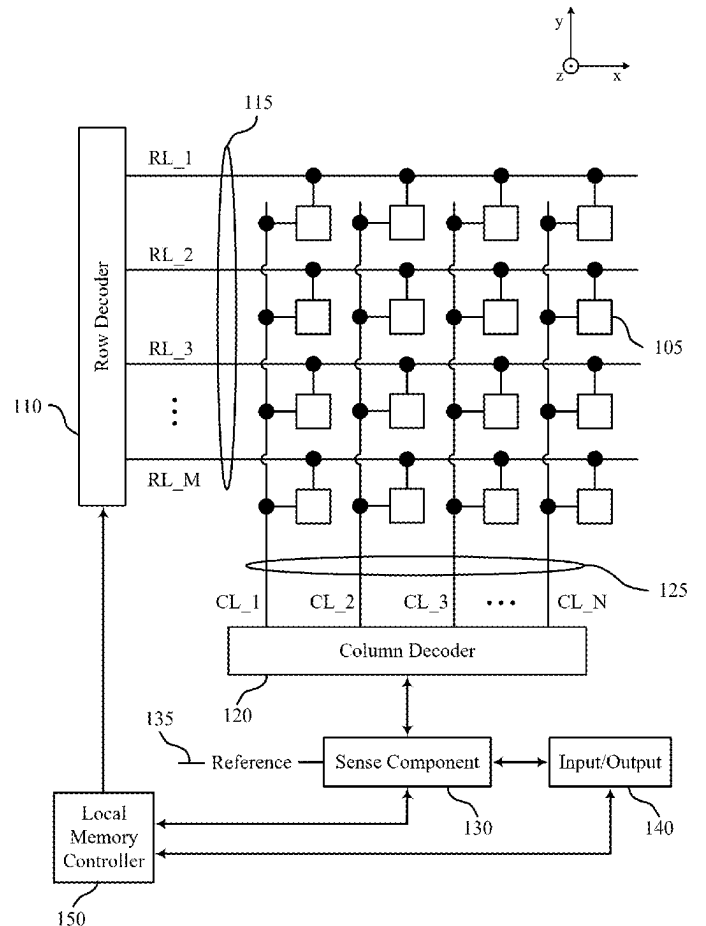
Methods, systems, and devices for shared decoder architecture for three-dimensional memory arrays are described. A memory device may include pillars coupled to an access line using two transistors positioned between the pillar and the access line. The gates of the two transistors may be coupled with respective gate lines coupled with circuitry configured to bias the gate line as part of an access operation for a memory cell coupled with the pillar. In some cases, the circuitry may be positioned between tiles of the memory device, at an end of one or more tiles of the memory device, between word line combs of a tile of the memory device, or a combination thereof.

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Related U.S. Application Data

(60) Continuation of application No. 18/622,033, filed on Mar. 29, 2024, now Pat. No. 12,260,907, which is a division of application No. 17/655,957, filed on Mar. 22, 2022, now Pat. No. 11,967,372.



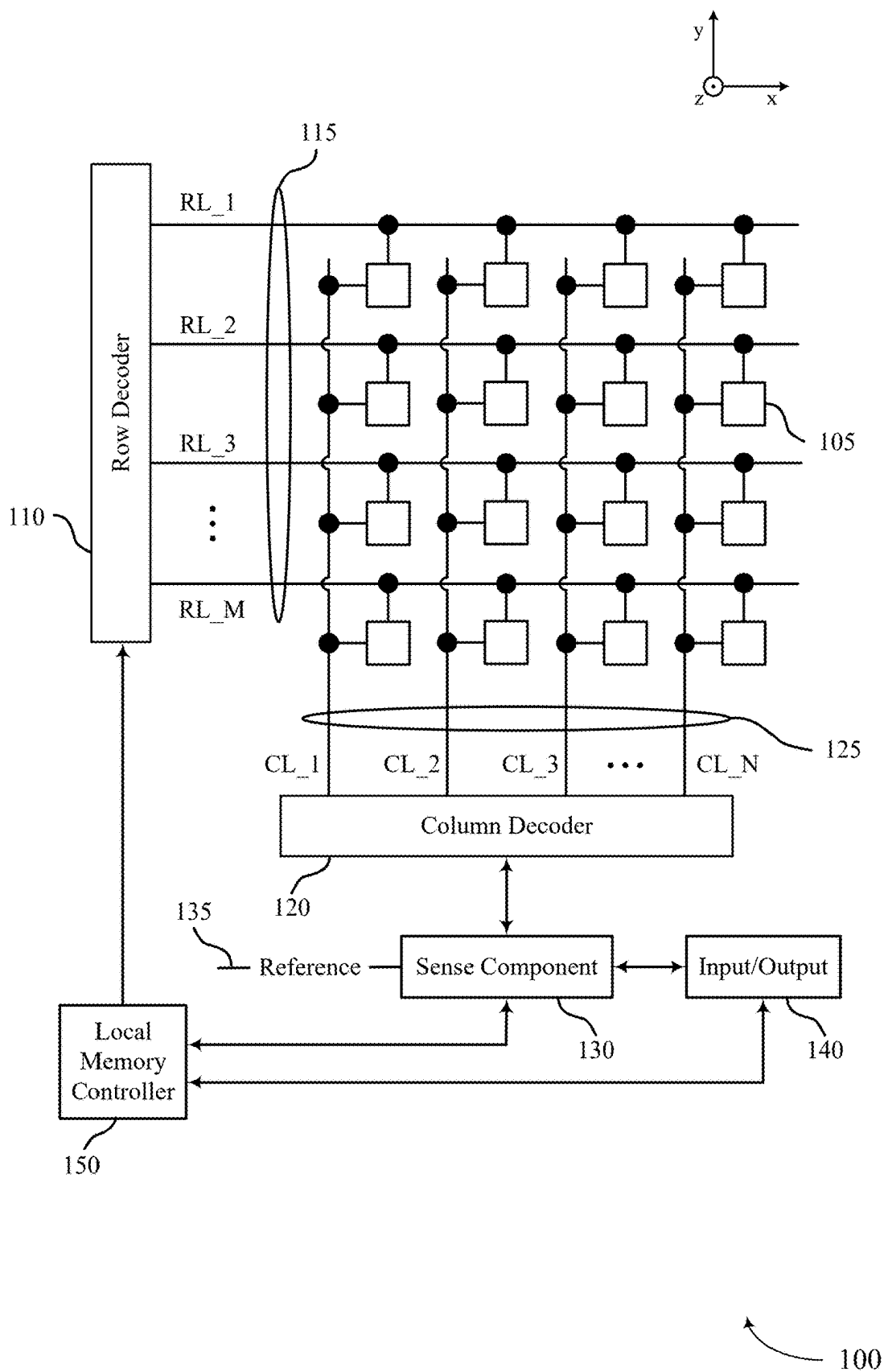
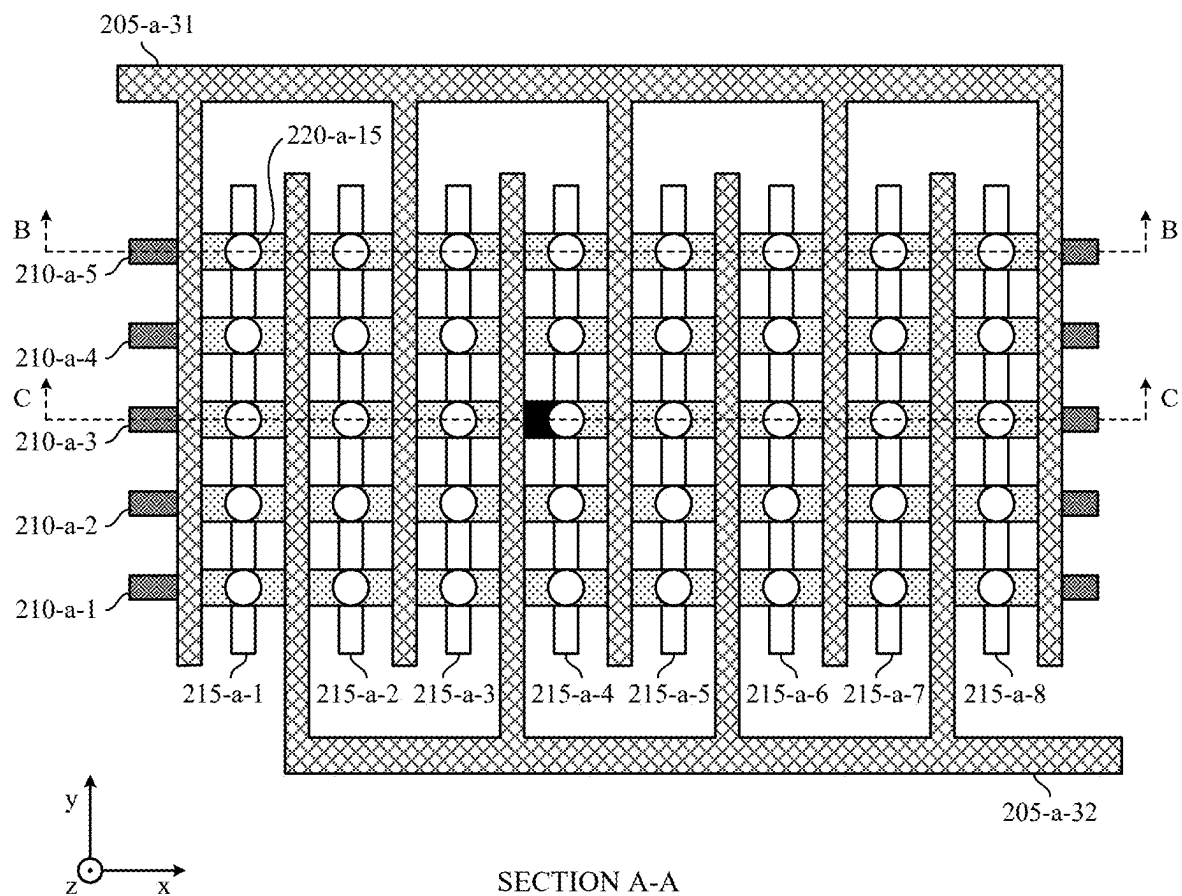


FIG. 1



Unselected Memory Cell 105

Selected Memory Cell 105-a

FIG. 2

200

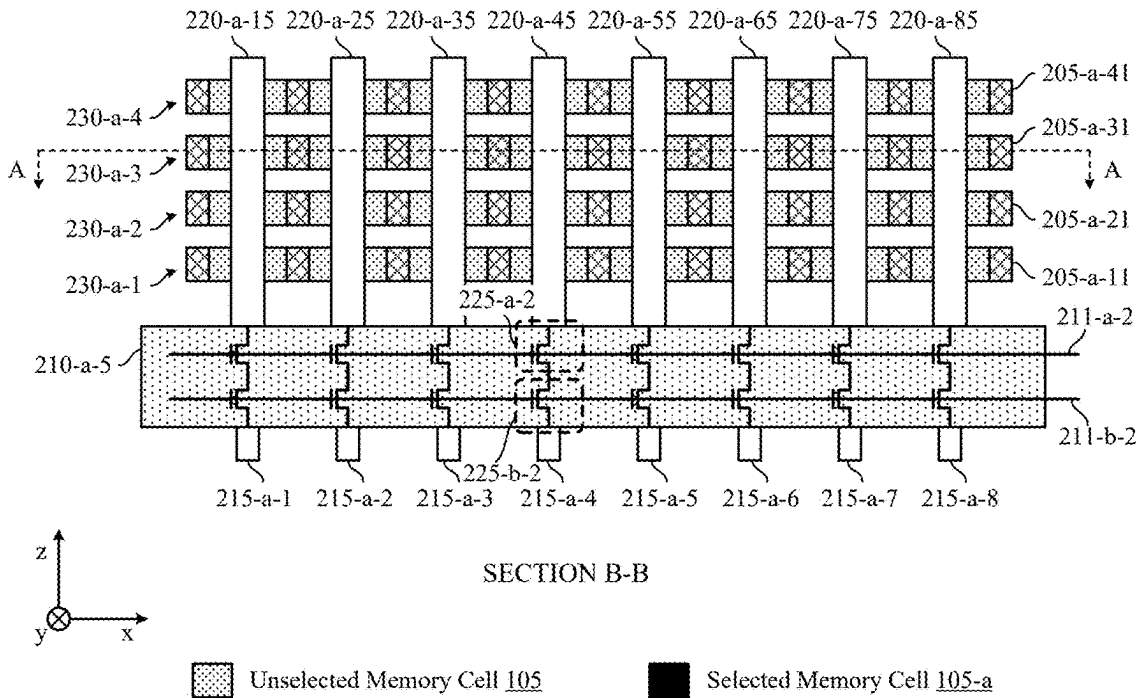


FIG. 3A

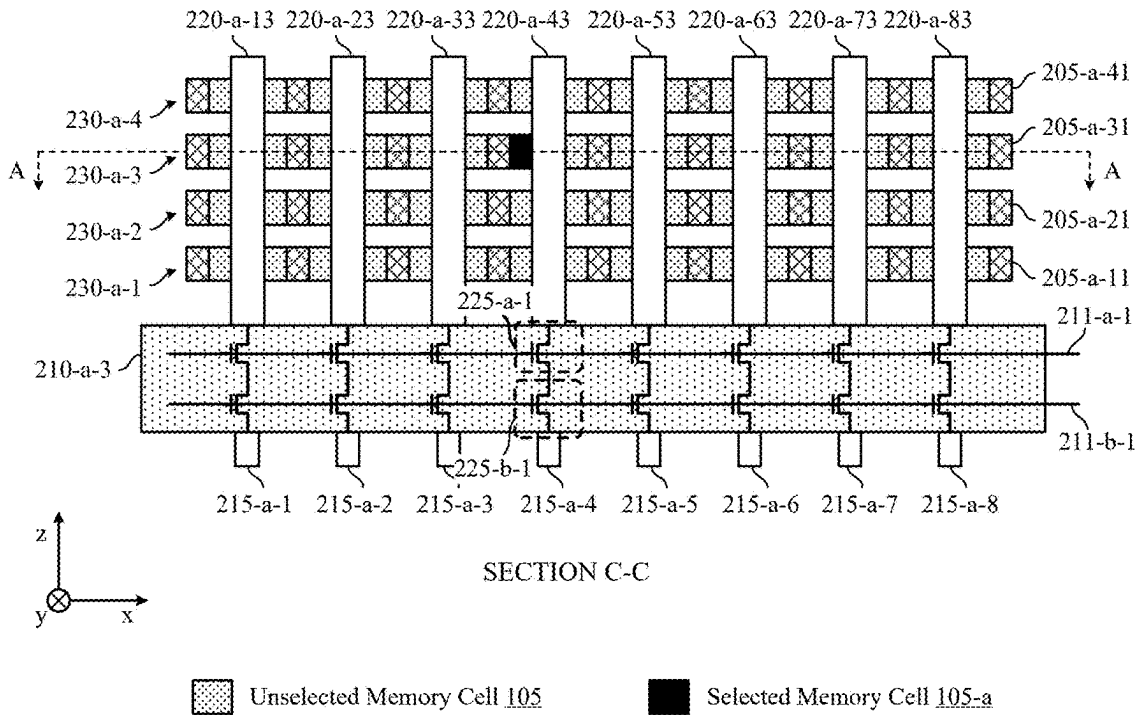


FIG. 3B

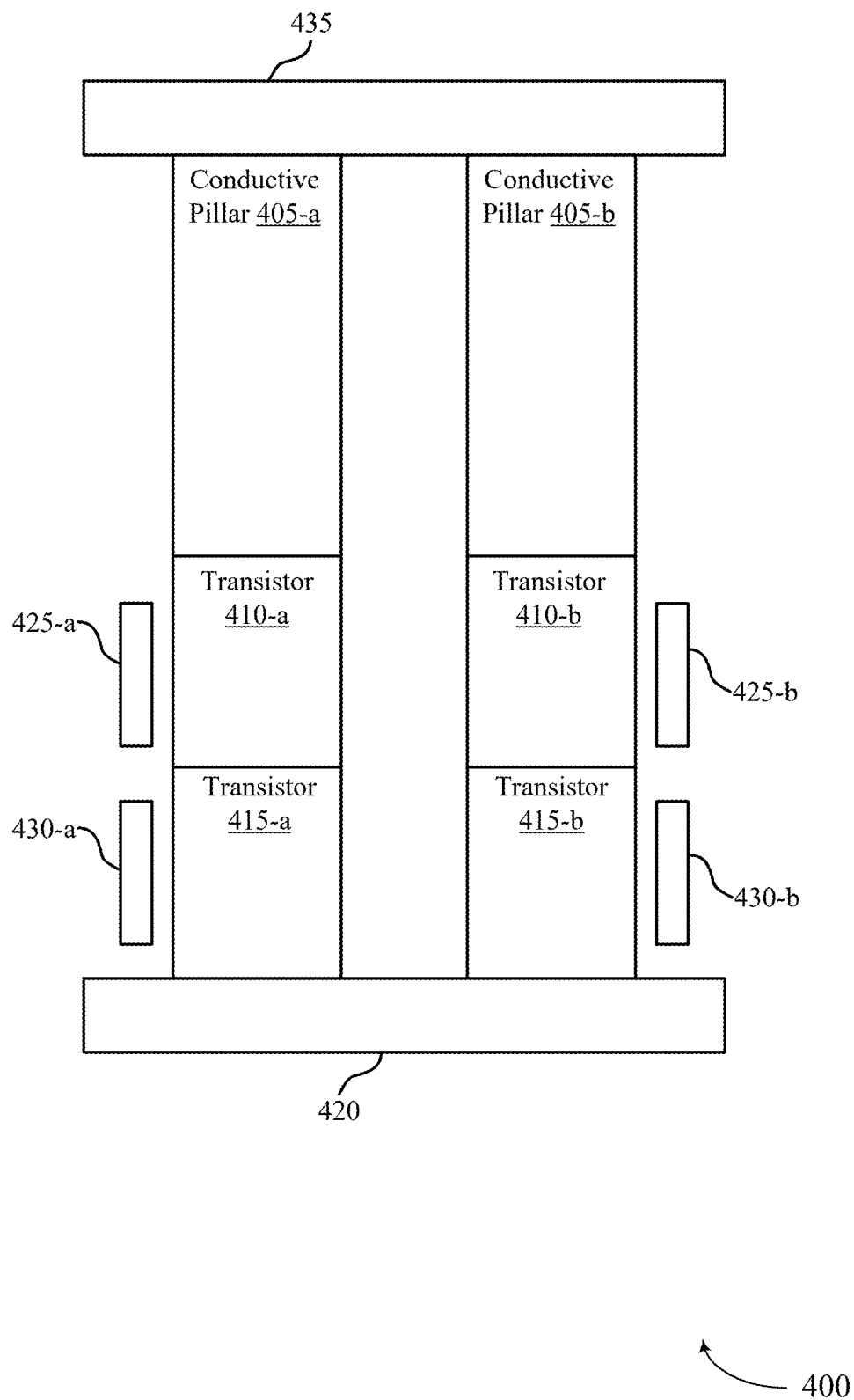


FIG. 4

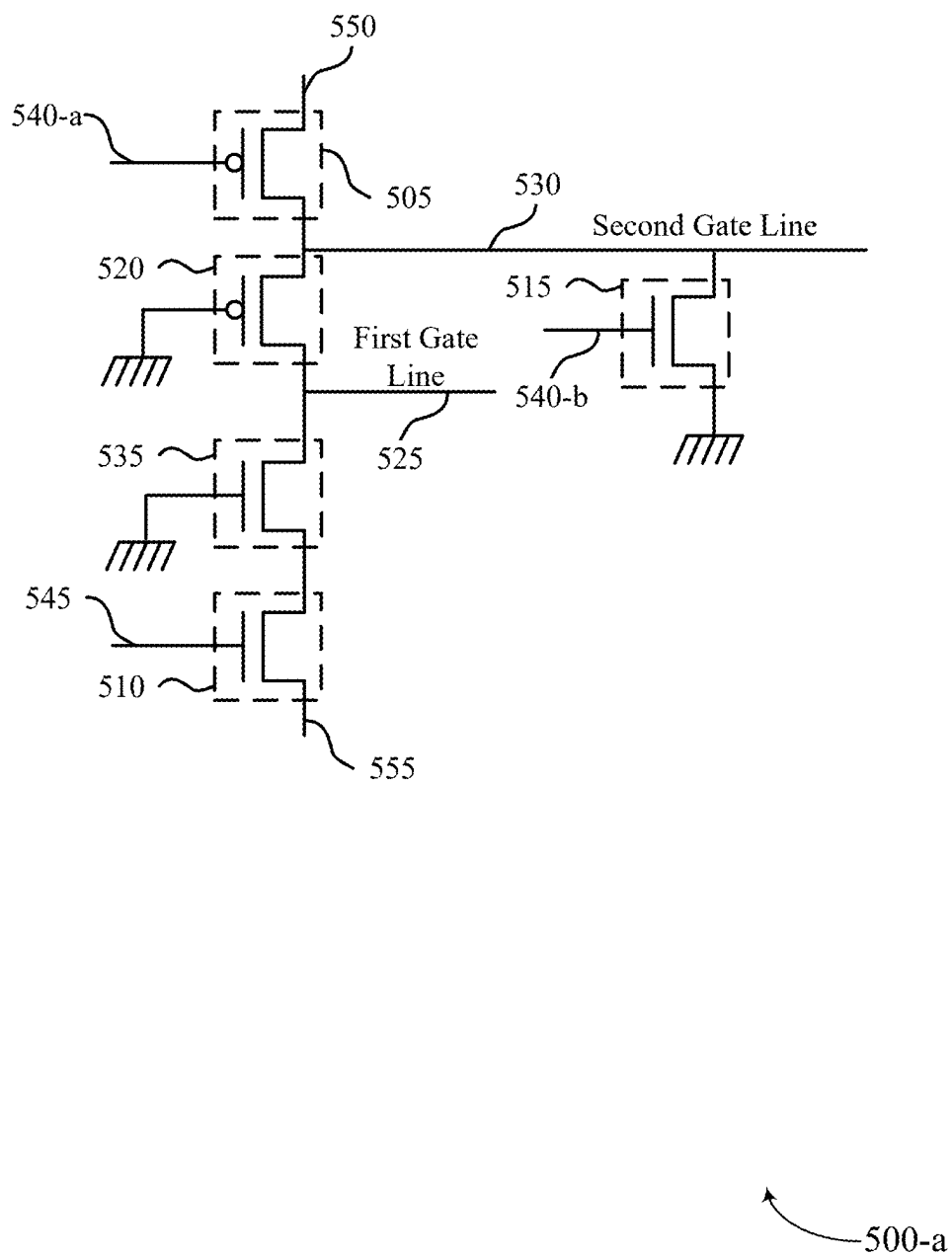
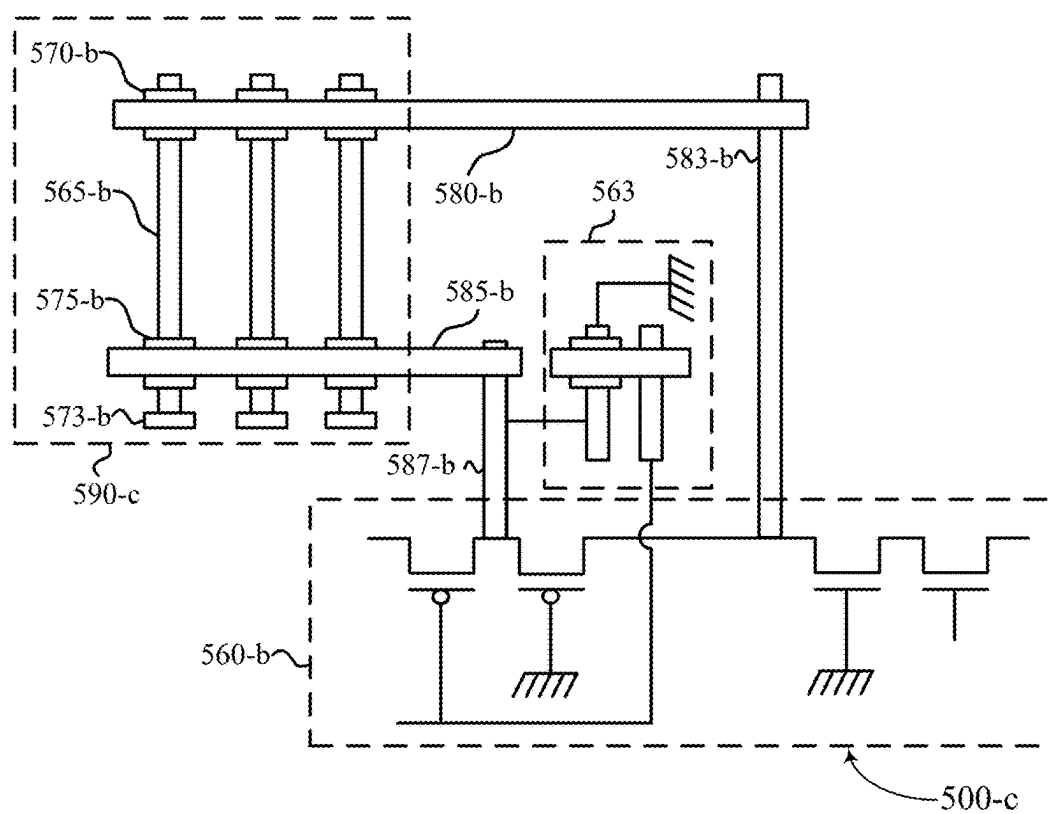
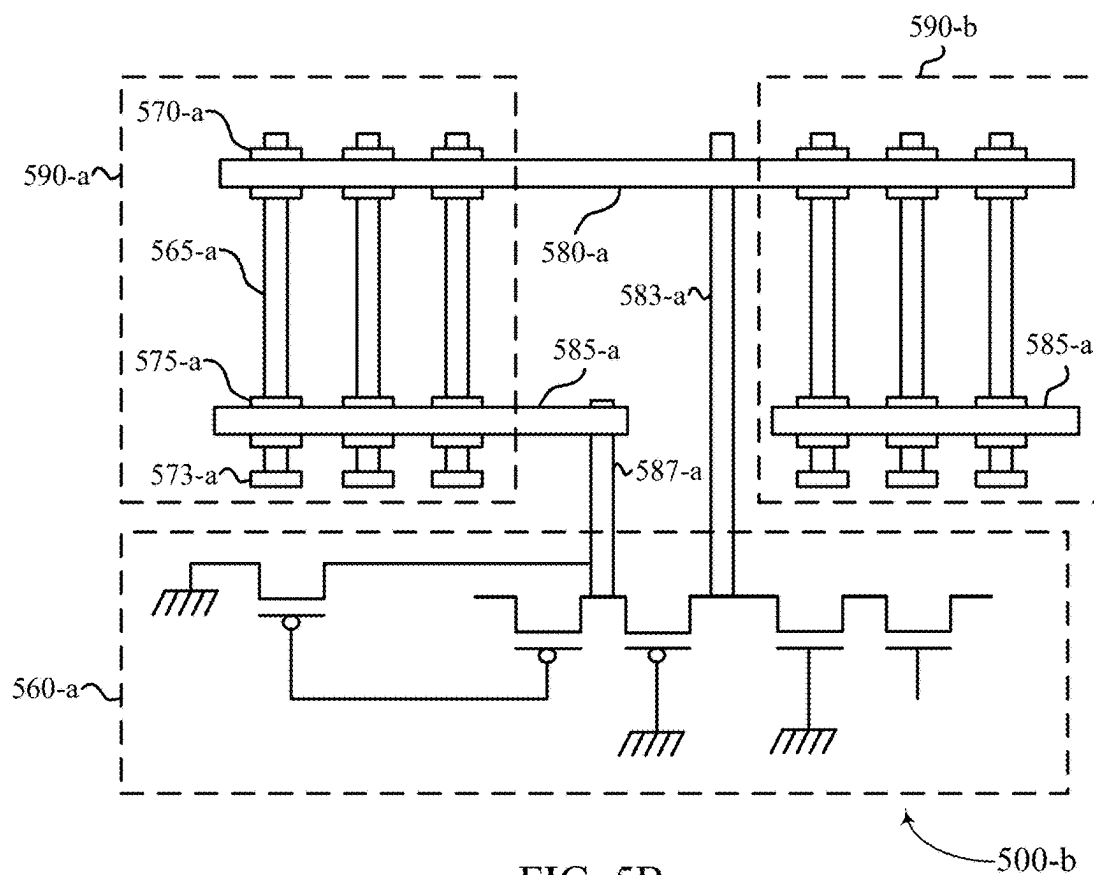


FIG. 5A



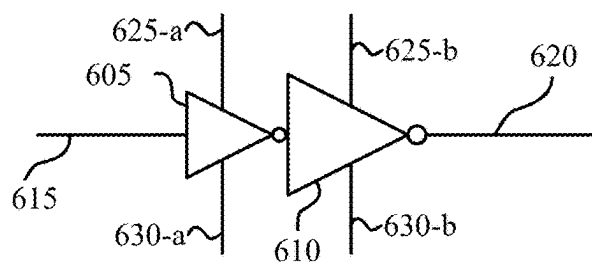


FIG. 6A

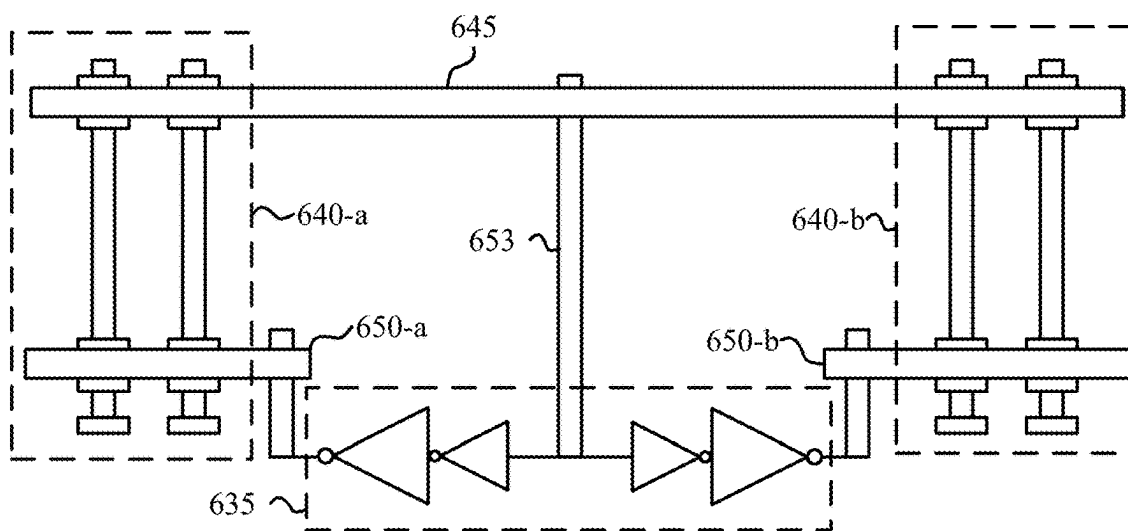


FIG. 6B

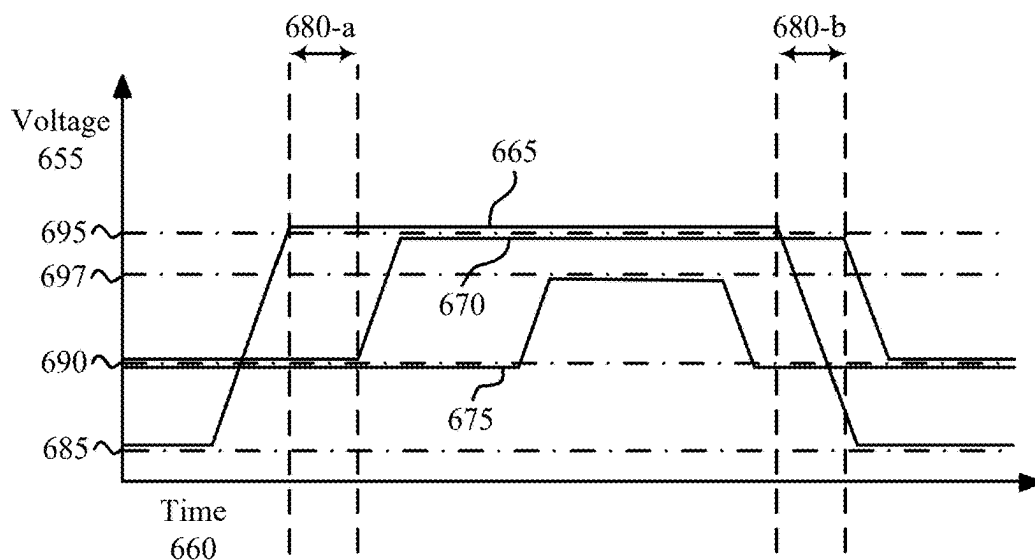


FIG. 6C

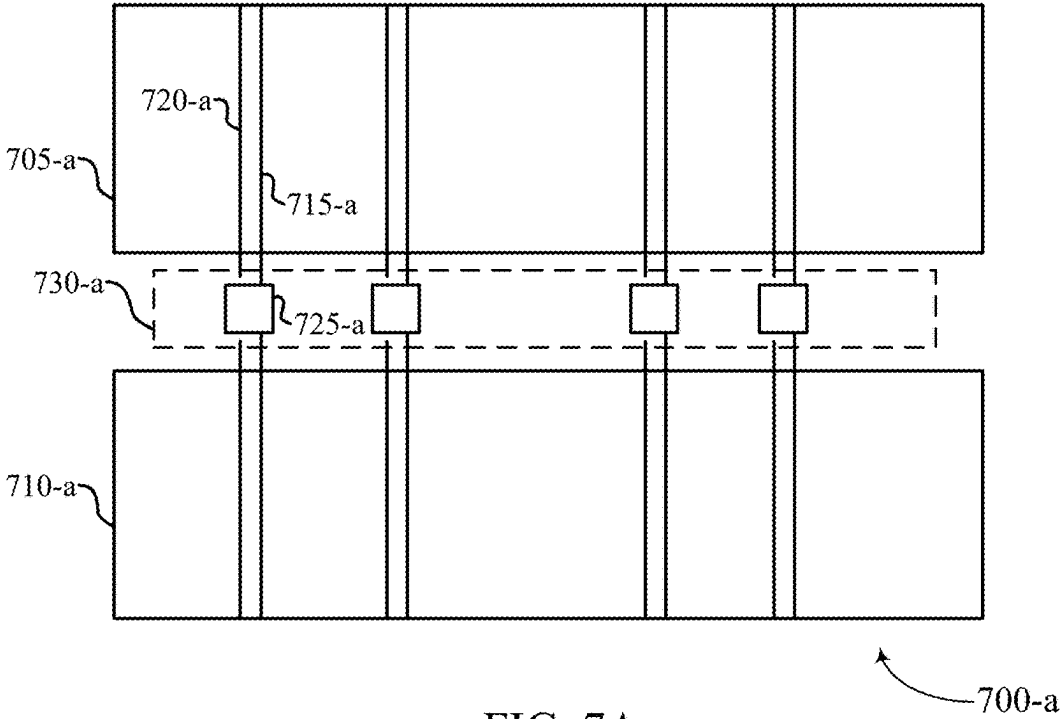


FIG. 7A

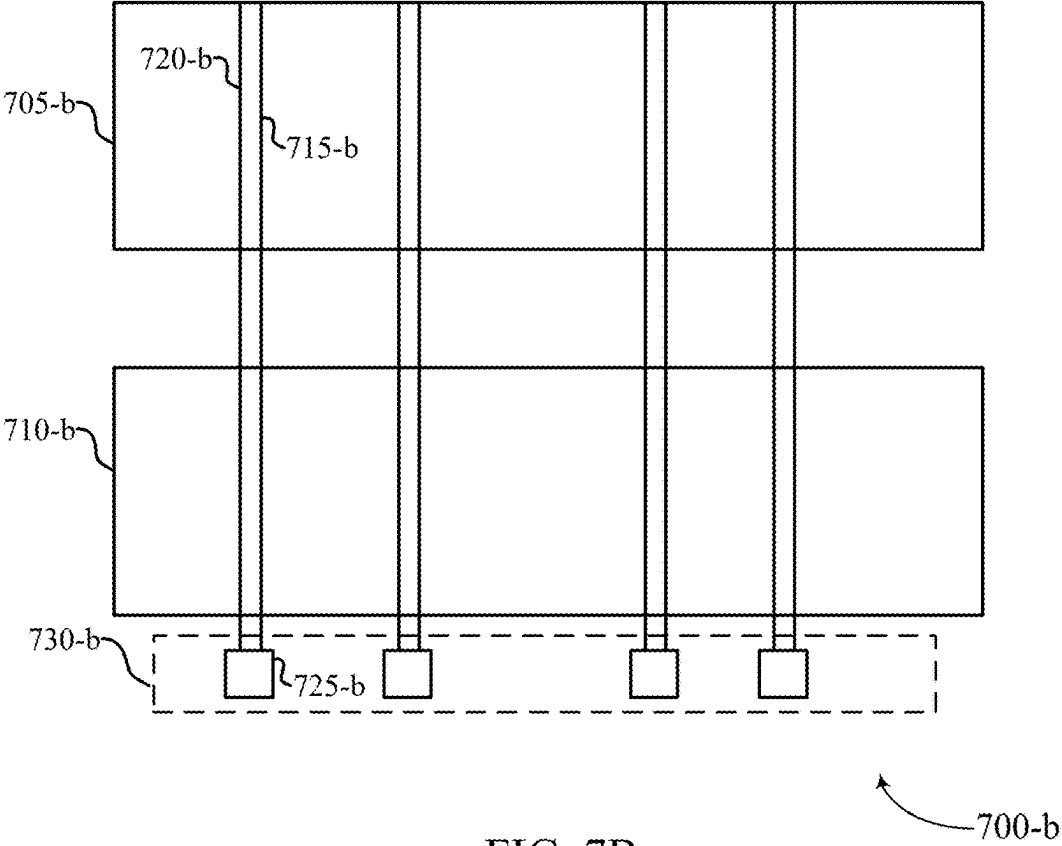
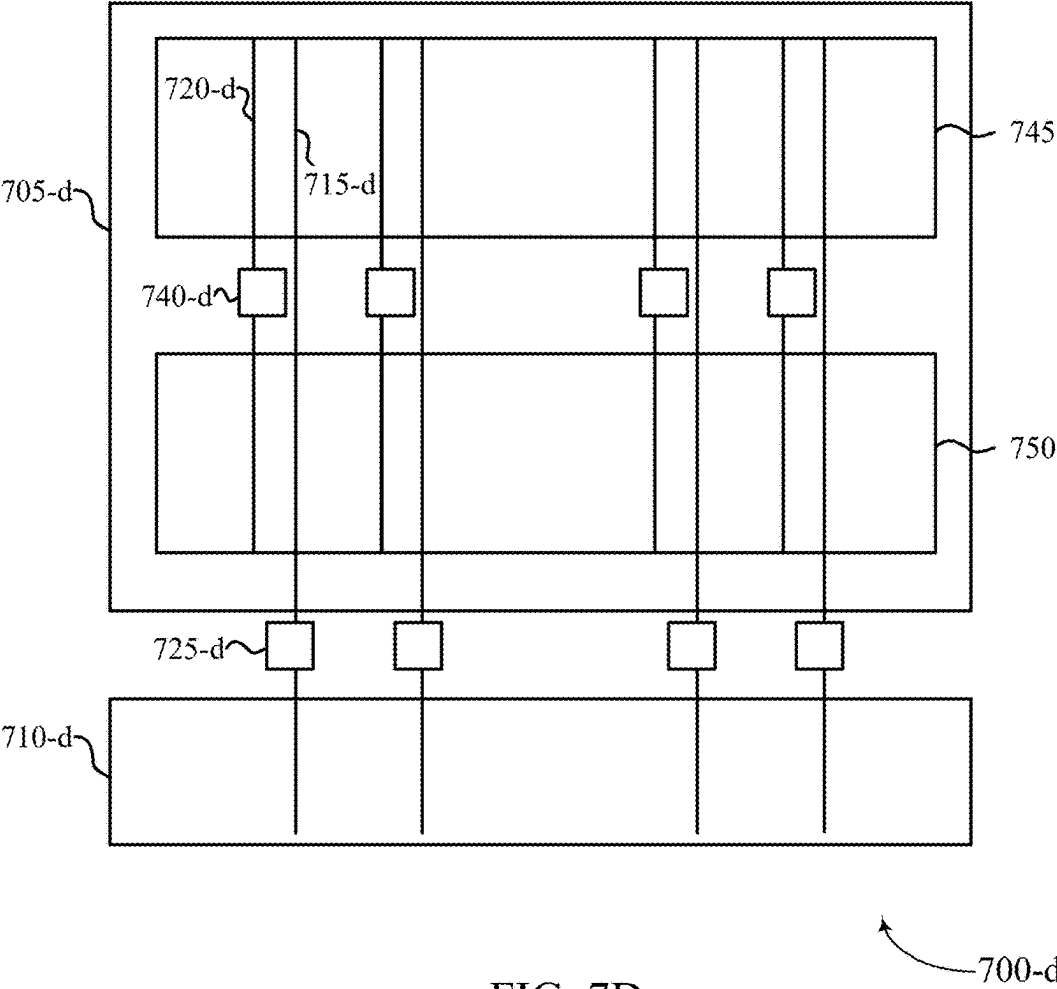
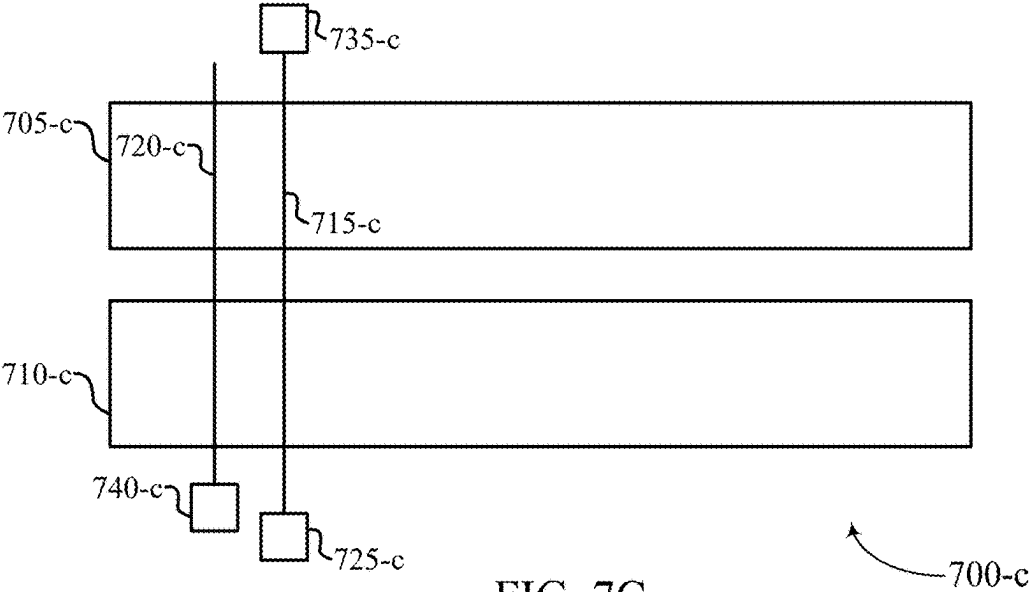
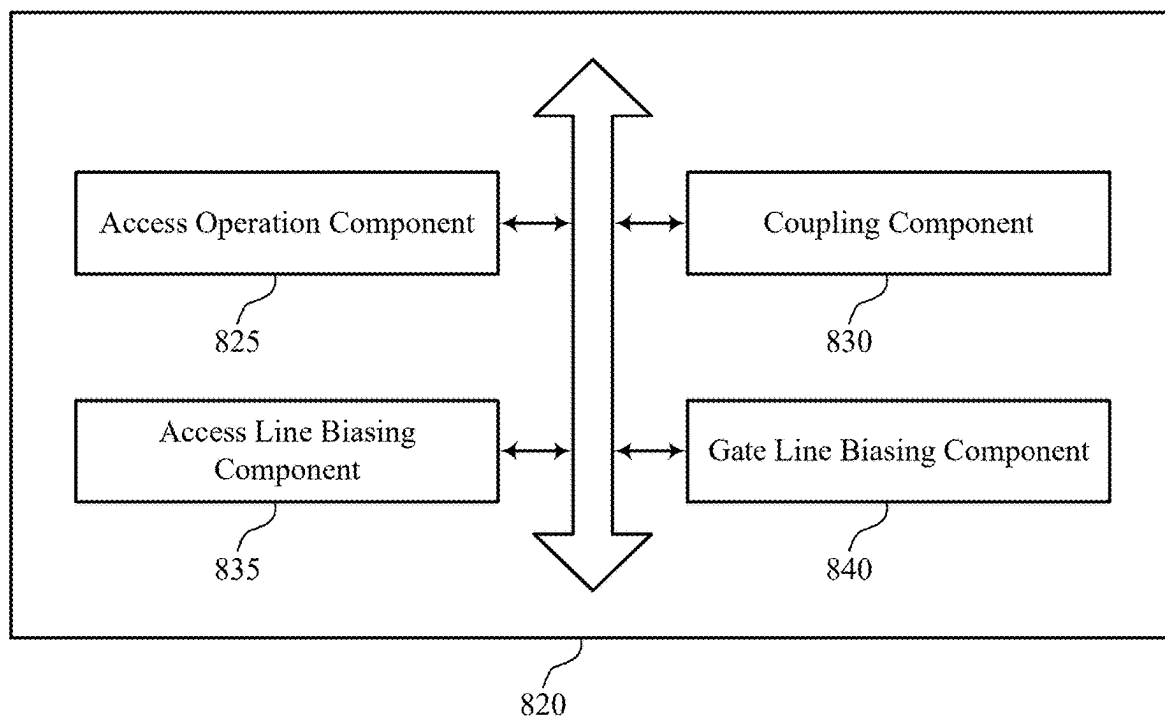


FIG. 7B





800

FIG. 8

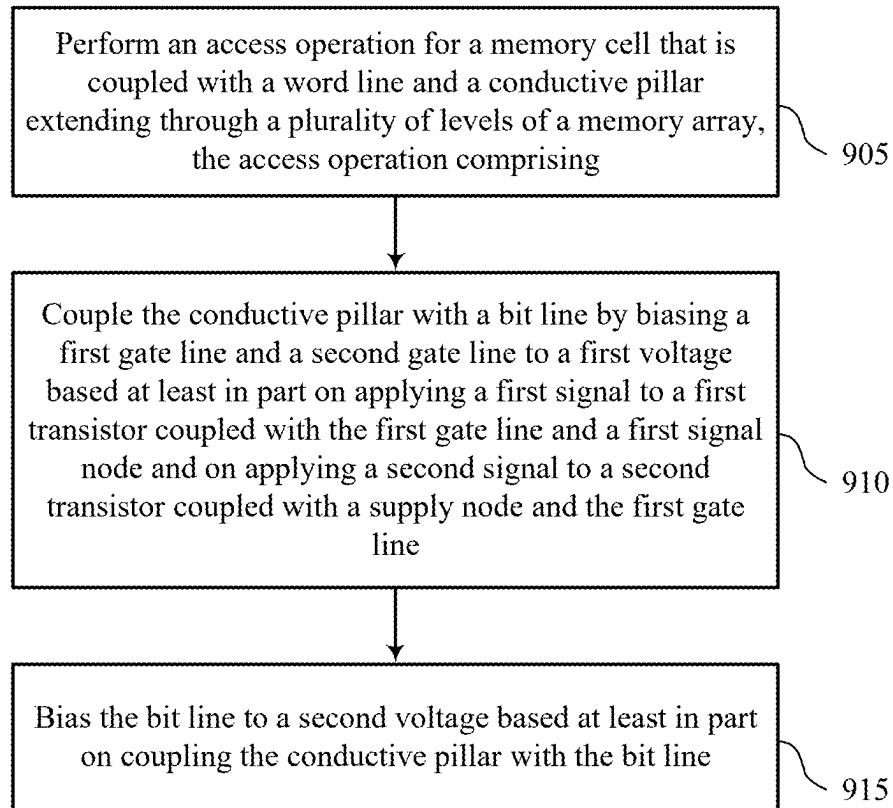


FIG. 9

SHARED DECODER ARCHITECTURE FOR THREE-DIMENSIONAL MEMORY ARRAYS

CROSS REFERENCE

[0001] The present Application for Patent is a continuation of U.S. patent application Ser. No. 18/622,033 by Laurent et al., entitled “SHARED DECODER ARCHITECTURE FOR THREE-DIMENSIONAL MEMORY ARRAYS,” filed Mar. 29, 2024, which is a divisional of U.S. patent application Ser. No. 17/655,957 by Laurent et al., entitled “SHARED DECODER ARCHITECTURE FOR THREE-DIMENSIONAL MEMORY ARRAYS,” filed Mar. 22, 2022, each of which is assigned to the assignee hereof, and each of which is expressly incorporated by reference in its entirety herein.

FIELD OF TECHNOLOGY

[0002] The following relates to one or more systems for memory, including shared decoder architecture for three-dimensional memory arrays.

BACKGROUND

[0003] Memory devices are widely used to store information in various electronic devices such as computers, user devices, wireless communication devices, cameras, digital displays, and the like. Information is stored by programming memory cells within a memory device to various states. For example, binary memory cells may be programmed to one of two supported states, often denoted by a logic 1 or a logic 0. In some examples, a single memory cell may support more than two states, any one of which may be stored. To access the stored information, a component may read (e.g., sense, detect, retrieve, identify, determine, evaluate) a stored state in the memory device. To store information, a component may write (e.g., program, set, assign) the state in the memory device.

[0004] Various types of memory devices and memory cells exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), static RAM (SRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), self-selecting memory, chalcogenide memory technologies, and others. Memory cells may be described in terms of volatile configurations or non-volatile configurations. Memory cells configured in a non-volatile configuration may maintain stored logic states for extended periods of time even in the absence of an external power source. Memory cells configured in a volatile configuration may lose stored states when disconnected from an external power source.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 illustrates an example of a memory array that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0006] FIG. 2 illustrates a top view of an example of a memory array that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0007] FIGS. 3A and 3B illustrate side views of an example of a memory array that supports shared decoder

architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0008] FIG. 4 illustrates an example of a memory array that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0009] FIGS. 5A, 5B, and 5C illustrate examples of circuitry that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0010] FIGS. 6A, 6B, and 6C illustrate examples of circuitry that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0011] FIG. 7A, 7B, 7C, and 7D illustrate examples of memory tiles that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0012] FIG. 8 shows a block diagram of a memory device that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

[0013] FIG. 9 shows a flowchart illustrating a method or methods that support shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein.

DETAILED DESCRIPTION

[0014] In some memory architectures, a memory device may include a memory array arranged in a vertical architecture, such as a three-dimensional architecture, that includes memory cells arranged according to different levels (e.g., layers, decks, planes, tiers). In some cases, a memory cell may be coupled with (e.g., coupled physically between, electrically between, or both) a word line and a pillar that extends vertically through the levels of the memory array. To access the memory cell, the memory device may include circuitry configured to bias the word line and the pillar to respective voltages such that a bias is applied across the memory cell. For example, a logic state may be written to the memory cell based on a current driven through the memory cell (e.g., an amount of current, a direction of current) as a result of the bias applied across the memory cell. In some cases, to bias a pillar as part of an access operation of a selected memory cell, the memory device may be configured to couple the pillar (e.g., a selected pillar) with a bit line by activating a transistor between the selected pillar and the bit line, while isolating other pillars (e.g., one or more unselected pillars) from the bit line by deactivating respective transistors between the one or more unselected pillars and the bit line, which may prevent or reduce a biasing across unselected memory cells.

[0015] A single transistor between a pillar and a bit line, however, may not fully isolate the bit line from the pillar when the pillar is unselected. In some examples, the bit line may be configured to be biased to wide swings of voltage (e.g., biased to both negative and positive voltages to change the polarity of the signal applied to memory cells). In such examples, a single transistor may not fully isolate the pillar from the bit line in all situations. In some examples, because the transistors between the pillars and the bit line may be subject to high electrical stress (e.g., many access operations), the transistors may degrade over time, which may lead to increased leakage current, memory cell failure,

decreased device performance, or a combination thereof. Accordingly, architectures to reduce improve isolation between the pillar and the bit line transistor stress are desired.

[0016] As described herein, a memory device may implement a multi-transistor architecture, such as a two-transistor architecture. For example, a memory array of the memory device may include pillars coupled with a bit line using two transistors positioned between the pillar and the bit line. As part of an access operation for a memory cell coupled with the pillar (e.g., the selected pillar), the memory device may be configured to bias the bit line to a first voltage and activate the two transistors using a second voltage to couple the pillar with the biased bit line. Additionally, the memory device may be configured to bias a gate of a first transistor and a gate of a second transistor coupling a second pillar to the bit line (e.g., an unselected pillar) to a third and fourth voltage, respectively, which may deactivate at least one of the first or second transistor during the access operation. Accordingly, the unselected pillar may remain isolated from the bit line during the access operation, and the transistors coupling the unselected pillar to the bit line may undergo less electrical stress.

[0017] In some cases, a memory device may include circuitry to bias a first gate line and a second gate line respectively coupled with the gate of the first transistor and the gate of the second transistor, such as a driver or decoder. For example, the circuitry may include one or more input signals which may be used to activate one or more pull up or pull down transistors to selectively bias the first gate line and the second gate. Additionally or alternatively, the circuitry may include a signal node coupled with the first gate, and a pair of inverters coupling the signal node with the second gate line. Accordingly, the first gate line and the second gate line may be selectively biased using a single signal applied to the signal node. In some examples, the circuitry may be positioned between tiles of the memory device, which may form a wall of vias coupled with the first gate line that may interrupt the second gate line. Additionally or alternatively, the circuitry may be positioned at an end of one or more tiles, such that the first gate line and the second gate line may be coupled with the circuitry without interruption. Thus, by implementing the multi-transistor architecture, such as the two-transistor architecture, a memory device may be implemented with reduced leakage through deactivated transistors, which may support reduced power consumption, increased accuracy of access operations, or any combination thereof, among other benefits.

[0018] Features of the disclosure are initially described in the context of memory devices and arrays with reference to FIGS. 1, 2, 3A, and 3B. Features of the disclosure are described in the context of memory arrays, circuits, and timing diagrams with reference to FIGS. 4-7D. These and other features of the disclosure are further illustrated by and described with reference to an apparatus diagram and flowcharts that relate to shared decoder architecture for three-dimensional memory arrays as described with reference to FIGS. 8-9.

[0019] FIG. 1 illustrates an example of a memory device 100 that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. In some examples, the memory device 100 may be referred to as or include a memory die, a memory chip, or an electronic memory apparatus. The memory

device 100 may be operable to provide locations to store information (e.g., physical memory addresses) that may be used by a system (e.g., a host device coupled with the memory device 100, for writing information, for reading information).

[0020] The memory device 100 may include one or more memory cells 105 that each may be programmable to store different logic states (e.g., a programmed one of a set of two or more possible states). For example, a memory cell 105 may be operable to store one bit of information at a time (e.g., a logic 0 or a logic 1). In some examples, a memory cell 105 (e.g., a multi-level memory cell 105) may be operable to store more than one bit of information at a time (e.g., a logic 00, logic 01, logic 10, a logic 11). In some examples, the memory cells 105 may be arranged in an array.

[0021] A memory cell 105 may store a logic state using a configurable material, which may be referred to as a memory element, a storage element, a memory storage element, a material element, a material memory element, a material portion, or a polarity-written material portion, among others. A configurable material of a memory cell 105 may refer to a chalcogenide-based storage component. For example, a chalcogenide storage element may be used in a phase change memory cell, a thresholding memory cell, or a self-selecting memory cell, among other architectures.

[0022] In some examples, the material of a memory cell 105 may include a chalcogenide material or other alloy including selenium (Se), tellurium (Te), arsenic (As), antimony (Sb), carbon (C), germanium (Ge), silicon (Si), or indium (In), or various combinations thereof. In some examples, a chalcogenide material having primarily selenium (Se), arsenic (As), and germanium (Ge) may be referred to as a SAG-alloy. In some examples, a SAG-alloy may also include silicon (Si) and such chalcogenide material may be referred to as SiSAG-alloy. In some examples, SAG-alloy may include silicon (Si) or indium (In) or a combination thereof and such chalcogenide materials may be referred to as SiSAG-alloy or InSAG-alloy, respectively, or a combination thereof. In some examples, the chalcogenide material may include additional elements such as hydrogen (H), oxygen (O), nitrogen (N), chlorine (Cl), or fluorine (F), each in atomic or molecular forms.

[0023] In some examples, a memory cell 105 may be an example of a phase change memory cell. In such examples, the material used in the memory cell 105 may be based on an alloy (such as the alloys listed above) and may be operated so as to change to different physical state (e.g., undergo a phase change) during normal operation of the memory cell 105. For example, a phase change memory cell 105 may be associated with a relatively disordered atomic configuration (e.g., a relatively amorphous state) and a relatively ordered atomic configuration (e.g., a relatively crystalline state). A relatively disordered atomic configuration may correspond to a first logic state (e.g., a RESET state, a logic 0) and a relatively ordered atomic configuration may correspond to a second logic state (e.g., a logic state different than the first logic state, a SET state, a logic 1).

[0024] In some examples (e.g., for thresholding memory cells 105, for self-selecting memory cells 105), some or all of the set of logic states supported by the memory cells 105 may be associated with a relatively disordered atomic configuration of a chalcogenide material (e.g., the material in an amorphous state may be operable to store different logic

states). In some examples, the storage element of a memory cell **105** may be an example of a self-selecting storage element. In such examples, the material used in the memory cell **105** may be based on an alloy (e.g., such as the alloys listed above) and may be operated so as to undergo a change to a different physical state during normal operation of the memory cell **105**. For example, a self-selecting or thresholding memory cell **105** may have a high threshold voltage state and a low threshold voltage state. A high threshold voltage state may correspond to a first logic state (e.g., a RESET state, a logic 0) and a low threshold voltage state may correspond to a second logic state (e.g., a logic state different than the first logic state, a SET state, a logic 1).

[0025] During a write operation (e.g., a programming operation) of a self-selecting or thresholding memory cell **105**, a polarity used for a write operation may influence (e.g., determine, set, program) a behavior or characteristic of the material of the memory cell **105**, such as a thresholding characteristic (e.g., a threshold voltage) of the material. A difference between thresholding characteristics of the material of the memory cell **105** for different logic states stored by the material of the memory cell **105** (e.g., a difference between threshold voltages when the material is storing a logic state '0' versus a logic state '1') may correspond to the read window of the memory cell **105**.

[0026] The memory device **100** may include access lines (e.g., row lines **115** each extending along an illustrative x-direction, column lines **125** each extending along an illustrative y-direction) arranged in a pattern, such as a grid-like pattern. Access lines may be formed with one or more conductive materials. In some examples, row lines **115**, or some portion thereof, may be referred to as word lines. In some examples, column lines **125**, or some portion thereof, may be referred to as digit lines or bit lines. References to access lines, or their analogues, are interchangeable without loss of understanding. Memory cells **105** may be positioned at intersections of access lines, such as row lines **115** and the column lines **125**. In some examples, memory cells **105** may also be arranged (e.g., addressed) along an illustrative z-direction, such as in an implementation of sets of memory cells **105** being located at different levels (e.g., layers, decks, planes, tiers) along the illustrative z-direction. In some examples, a memory device **100** that includes memory cells **105** at different levels may be supported by a different configuration of access lines, decoders, and other supporting circuitry than shown.

[0027] Operations such as read operations and write operations may be performed on the memory cells **105** by activating access lines such as one or more of a row line **115** or a column line **125**, among other access lines associated with alternative configurations. For example, by activating a row line **115** and a column line **125** (e.g., applying a voltage to the row line **115** or the column line **125**), a memory cell **105** may be accessed in accordance with their intersection. An intersection of a row line **115** and a column line **125**, among other access lines, in various two-dimensional or three-dimensional configuration may be referred to as an address of a memory cell **105**. In some examples, an access line may be a conductive line coupled with a memory cell **105** and may be used to perform access operations on the memory cell **105**. In some examples, the memory device **100** may perform operations responsive to commands, which may be issued by a host device coupled with the memory

device **100** or may be generated by the memory device **100** (e.g., by a local memory controller **150**).

[0028] Accessing the memory cells **105** may be controlled through one or more decoders, such as a row decoder **110** or a column decoder **120**, among other examples. For example, a row decoder **110** may receive a row address from the local memory controller **150** and activate a row line **115** based on the received row address. A column decoder **120** may receive a column address from the local memory controller **150** and may activate a column line **125** based on the received column address.

[0029] The sense component **130** may be operable to detect a state (e.g., a material state, a resistance state, a threshold state) of a memory cell **105** and determine a logic state of the memory cell **105** based on the detected state. The sense component **130** may include one or more sense amplifiers to convert (e.g., amplify) a signal resulting from accessing the memory cell **105** (e.g., a signal of a column line **125** or other access line). The sense component **130** may compare a signal detected from the memory cell **105** to a reference **135** (e.g., a reference voltage, a reference charge, a reference current). The detected logic state of the memory cell **105** may be provided as an output of the sense component **130** (e.g., to an input/output component **140**), and may indicate the detected logic state to another component of the memory device **100** or to a host device coupled with the memory device **100**.

[0030] The local memory controller **150** may control the accessing of memory cells **105** through the various components (e.g., a row decoder **110**, a column decoder **120**, a sense component **130**, among other components). In some examples, one or more of a row decoder **110**, a column decoder **120**, and a sense component **130** may be co-located with the local memory controller **150**. The local memory controller **150** may be operable to receive information (e.g., commands, data) from one or more different controllers (e.g., an external memory controller associated with a host device, another controller associated with the memory device **100**), translate the information into a signaling that can be used by the memory device **100**, perform one or more operations on the memory cells **105** and communicate data from the memory device **100** to a host device based on performing the one or more operations. The local memory controller **150** may generate row address signals and column address signals to activate access lines such as a target row line **115** and a target column line **125**. The local memory controller **150** also may generate and control various signals (e.g., voltages, currents) used during the operation of the memory device **100**. In general, the amplitude, the shape, or the duration of an applied signal discussed herein may be varied and may be different for the various operations discussed in operating the memory device **100**.

[0031] The local memory controller **150** may be operable to perform one or more access operations on one or more memory cells **105** of the memory device **100**. Examples of access operations may include a write operation, a read operation, a refresh operation, a precharge operation, or an activate operation, among others. In some examples, access operations may be performed by or otherwise coordinated by the local memory controller **150** in response to access commands (e.g., from a host device). The local memory controller **150** may be operable to perform other access operations not listed here or other operations related to the

operating of the memory device **100** that are not directly related to accessing the memory cells **105**.

[0032] In some cases, a memory device **100** may implement a multi-transistor architecture, such as a two-transistor architecture. For example, the memory device **100** may include pillars coupled to an access line, such as a column line **125**, using two transistors positioned between the pillar and the column line **125**. As part of an access operation for a memory cell **105** coupled with the pillar (e.g., the selected pillar), the memory device **100** may be configured to bias the column line **125** to a first voltage and activate the two transistors using a second voltage to couple the pillar with the biased column line **125**. Additionally, the memory device **100** may include circuitry, such as a set of pull up transistors and pull down transistors, to bias a gate of a first transistor and a gate of a second transistor coupling a second pillar to column line **125** (e.g., an unselected pillar) to a third and fourth voltage, respectively, which may deactivate at least one of the first or second transistor during the access operation. In some cases, the circuitry may be included in the row decoder **110**, the column decoder **120**, or both. Accordingly, the unselected pillar may remain at least partially isolated from the column line **125** during the access operation, and the transistors coupling the unselected pillar to the column line **125** may undergo less electrical stress.

[0033] The memory device **100** may include any quantity of non-transitory computer readable media that support shared decoder architecture for three-dimensional memory arrays. For example, a local memory controller **150**, a row decoder **110**, a column decoder **120**, a sense component **130**, or an input/output component **140**, or any combination thereof may include or may access one or more non-transitory computer readable media storing instructions (e.g., firmware) for performing the functions ascribed herein to the memory device **100**. For example, such instructions, if executed by the memory device **100**, may cause the memory device **100** to perform one or more associated functions as described herein.

[0034] FIGS. 2, 3A, and 3B illustrate an example of a memory array **200** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array **200** may be included in a memory device **100**, and illustrates an example of a three-dimensional arrangement of memory cells **105** that may be accessed by various conductive structures (e.g., access lines). FIG. 2 illustrates a top section view (e.g., SECTION A-A) of the memory array **200** relative to a cut plane A-A as shown in FIGS. 3A and 3B. FIG. 3A illustrates a side section view (e.g., SECTION B-B) of the memory array **200** relative to a cut plane B-B as shown in FIG. 2. FIG. 3B illustrates a side section view (e.g., SECTION C-C) of the memory array **200** relative to a cut plane C-C as shown in FIG. 2. The section views may be examples of cross-sectional views of the memory array **200** with some aspects (e.g., dielectric structures) removed for clarity. Elements of the memory array **200** may be described relative to an x-direction, a y-direction, and a z-direction, as illustrated in each of FIGS. 2, 3A, and 3B. Although some elements included in FIGS. 2, 3A, and 3B are labeled with a numeric indicator, other corresponding elements are not labeled, although they are the same or would be understood to be similar, in an effort to increase visibility and clarity of the depicted features. Further, although some quantities of repeated elements are shown in the illustrative example of

memory array **200**, techniques in accordance with examples as described herein may be applicable to any quantity of such elements, or ratios of quantities between one repeated element and another.

[0035] In the example of memory array **200**, memory cells **105** and word lines **205** may be distributed along the z-direction according to levels **230** (e.g., decks, layers, planes, as illustrated in FIGS. 3A and 3B). In some examples, the z-direction may be orthogonal to a substrate (not shown) of the memory array **200**, which may be below the illustrated structures along the z-direction. Although the illustrative example of memory array **200** includes four levels **230**, a memory array **200** in accordance with examples as disclosed herein may include any quantity of one or more levels **230** (e.g., 64 levels, 128 levels) along the z-direction.

[0036] Each word line **205** may be an example of a portion of an access line that is formed by one or more conductive materials (e.g., one or more metal portions, one or more metal alloy portions). As illustrated, a word line **205** may be formed in a comb structure, including portions (e.g., projections, tines) extending along the y-direction through gaps (e.g., alternating gaps) between pillars **220**. For example, as illustrated, the memory array **200**, may include two word lines **205** per level **230** (e.g., according to odd word lines **205-a-n1** and even word lines **205-a-n2** for a given level, *n*), where such word lines **205** of the same level **230** may be described as being interleaved (e.g., with portions of an odd word line **205-a-n1** projecting along the y-direction between portions of an even word line **205-a-n2**, and vice versa). In some examples, an odd word line **205** (e.g., of a level **230**) may be associated with a first memory cell **105** on a first side (e.g., along the x-direction) of a given pillar **220** and an even word line (e.g., of the same level **230**) may be associated with a second memory cell **105** on a second side (e.g., along the x-direction, opposite the first memory cell **105**) of the given pillar **220**. Thus, in some examples, memory cells **105** of a given level **230** may be addressed (e.g., selected, activated) in accordance with an even word line **205** or an odd word line **205**.

[0037] Each pillar **220** may be an example of a portion of an access line (e.g., a conductive pillar portion) that is formed by one or more conductive materials (e.g., one or more metal portions, one or more metal alloy portions). As illustrated, the pillars **220** may be arranged in a two-dimensional array (e.g., in an xy-plane) having a first quantity of pillars **220** along a first direction (e.g., eight pillars along the x-direction, eight rows of pillars), and having a second quantity of pillars **220** along a second direction (e.g., five pillars along the y-direction, five columns of pillars). Although the illustrative example of memory array **200** includes a two-dimensional arrangement of eight pillars **220** along the x-direction and five pillars **220** along the y-direction, a memory array **200** in accordance with examples as disclosed herein may include any quantity of pillars **220** along the x-direction and any quantity of pillars **220** along the y-direction. Further, as illustrated, each pillar **220** may be coupled with a respective set of memory cells **105** (e.g., along the z-direction, one or more memory cells **105** for each level **230**). A pillar **220** may have a cross-sectional area in an xy-plane that extends along the z-direction. Although illustrated with a circular cross-sectional area in the xy-plane, a pillar **220** may be formed with

a different shape, such as having an elliptical, square, rectangular, polygonal, or other cross-sectional area in an xy-plane.

[0038] The memory cells **105** each may include a chalcogenide material. In some examples, the memory cells **105** may be examples of thresholding memory cells. Each memory cell **105** may be accessed (e.g., addressed, selected) according to an intersection between a word line **205** (e.g., a level selection, which may include an even or odd selection within a level **230**) and a pillar **220**. For example, as illustrated, a selected memory cell **105-a** of the level **230-a-3** may be accessed according to an intersection between the pillar **220-a-43** and the word line **205-a-32**.

[0039] A memory cell **105** may be accessed (e.g., written to, read from) by applying an access bias (e.g., an access voltage, V_{access} , which may be a positive voltage or a negative voltage) across the memory cell **105**. In some examples, an access bias may be applied by biasing a selected word line **205** with a first voltage (e.g., $V_{access}/2$) and by biasing a selected pillar **220** with a second voltage (e.g., $-V_{access}/2$), which may have an opposite sign relative to the first voltage. Regarding the selected memory cell **105-a**, a corresponding access bias (e.g., the first voltage) may be applied to the word line **205-a-32**, while other unselected word lines **205** may be grounded (e.g., biased to 0V). In some examples, a word line bias may be provided by a word line driver (not shown) coupled with one or more of the word lines **205**.

[0040] To apply a corresponding access bias (e.g., the second voltage) to a pillar **220**, the pillars **220** may be configured to be selectively coupled with a bit line **215** (e.g., a digit line, a column line, an access line extending along the y-direction) via a pair of transistors coupled between (e.g., physically, electrically) the pillar **220** and the bit line **215**. The pair of transistors for each pillar may include a first transistor **225-a** and a second transistor **225-b**. In some examples, the transistors **225-a** and **225-b** may be vertical transistors (e.g., transistors having a channel along the z-direction, transistors having a semiconductor junction along the z-direction), which may be formed above the substrate of the memory array **200** using various techniques (e.g., thin film techniques). In some examples, a selected pillar **220**, a selected bit line **215**, or a combination thereof may be an example of a selected column line **125** described with reference to FIG. 1 (e.g., a bit line).

[0041] The transistors **225-a** and **225-b** (e.g., a channel portion of the transistors **225-a** and **225-b**) may be activated by a set of gate lines **210** (e.g., activation lines, selection lines, an access line extending along the x-direction) coupled with respective gates of a set of the transistors **225-a** and **225-b** (e.g., a set along the x-direction). In some cases, the set of gates line **210** may include two gate lines, a first gate line **211-a** and a second gate line **211-b**. The first gate line **211-a** may be coupled with gates of the first transistors **225-a** of the transistor pairs and may be configured to activate or deactivate the first transistors **225-a** based on the voltage applied to the first gate line **211-a**. The second gate line **211-b** may be coupled with the gates of the second transistors **225-b** of the transistor pairs and may be configured to activate or deactivate the second transistors **225-b** based on the voltage applied to the second gate line **211-b**. Each of the pillars **220** may have a first end (e.g., towards the negative z-direction, a bottom end) configured for coupling with an access line (e.g., a bit line **215**). In some examples,

the set of gate lines **210** (e.g., gate lines **211-a** and **211-b**), the transistors **225-a** and **225-b**, may be considered to be components of a row decoder **110** (e.g., as pillar decoder components). In some examples, the selection of (e.g., biasing of) pillars **220**, or bit lines **215**, or various combinations thereof, may be supported by a column decoder **120**, or a sense component **130**, or both.

[0042] To apply the corresponding access bias (e.g., $-V_{access}/2$) to the pillar **220-a-43**, the bit line **215-a-4** may be biased with the access bias, and the set of gate lines **210-a-3** may be grounded (e.g., biased to 0V) or otherwise biased with an activation voltage. In an example where the transistors **225-a-1** and **225-b** are n-type transistors, the set of gate lines **210-a-3** (e.g., gate lines **211-a-1** and **211-b-1**) may be biased with a voltage that is relatively higher than the bit line **215-a-4** and may activate the transistors **225-a** and **225-b** (e.g., cause the transistors **225-a** and **225-b** to operate in a conducting state). Thereby coupling the pillar **220-a-43** with the bit line **215-a-4** and biasing the pillar **220-a-43** with the associated access bias. However, the transistors **225-a** and **225-b** may include different channel types, or may be operated in accordance with different biasing schemes, to support various access operations.

[0043] In some examples, unselected pillars **220** of the memory array **200** may be electrically floating when the transistors **225-a-1** and **225-b-1** for the selected pillar (e.g., pillar **220-a-45**) are activated, or may be coupled with another voltage source (e.g., grounded, via a high-resistance path, via a leakage path) to avoid a voltage drift of the unselected pillars **220**. For example, a different voltage (e.g., ground voltage) being applied to the set of gate lines **210-a-3** (e.g., gate lines **211-a-1** and **211-b-1**) may not activate the transistors **225-a-1** and **225-b-1** coupled with the set of gate lines **210-a-3** (e.g., gate lines **211-a-1** and **211-b-1**), because the different voltage of the set of gate lines **210-a-3** (e.g., gate lines **211-a-1** and **211-b-1**) may not be different enough than the voltage of the other bit lines **215**.

[0044] The set of gate lines **210-a-5** may be biased to the different voltage (e.g., gate lines **211-a-2** and **211-b-2**) to isolate the conductive pillar **220-a-45** from the bit line **215-a-4** that is biased to access a memory cell coupled with the selected conductive pillar **220-a-43**. In some examples, the gate line **211-a-2** may be biased to a different voltage than the gate line **211-b-2** to ensure that the unselected conductive pillar **220-a-45** is isolated from the bit line **215-a-4** for different biases of the bit line **215-a-4**. Different access operations may cause the bit line **215-a-4** to be biased to different voltages. Using two transistors (e.g., transistors **225-a-2** and **225-b-2**) biased to different gate voltages may increase a likelihood of isolation between the unselected conductive pillar and the bit line and reduce stress on the transistors. Thus, at least one of the transistors **225-a** and **225-b** coupled with an unselected pillar may be deactivated (e.g., operating in a non-conductive state), thereby isolating the voltage of the bit line **215-a-4** from the pillar **220-a-45**, among other pillars **220**.

[0045] In a write operation, a memory cell **105** may be written to by applying a write bias (e.g., where $V_{access}=V_{write}$, which may be a positive voltage or a negative voltage) across the memory cell **105**. In some examples, a polarity of a write bias may influence (e.g., determine, set, program) a behavior or characteristic of the material of the memory cell **105**, such as the threshold voltage of the material. For example, applying a write bias with a first

polarity may set the material of the memory cell **105** with a first threshold voltage, which may be associated with storing a logic 0. Further, applying a write bias with a second polarity (e.g., opposite the first polarity) may set the material of the memory cell with a second threshold voltage, which may be associated with storing a logic 1. A difference between threshold voltages of the material of the memory cell **105** for different logic states stored by the material of the memory cell **105** (e.g., a difference between threshold voltages when the material is storing a logic state '0' versus a logic state '1') may correspond to the read window of the memory cell **105**.

[0046] In a read operation, a memory cell **105** may be read from by applying a read bias (e.g., where $V_{access} = V_{read}$, which may be a positive voltage or a negative voltage) across the memory cell **105**. In some examples, a logic state of the memory cell **105** may be evaluated based on whether the memory cell **105** thresholds in the presence of the applied read bias. For example, such a read bias may cause a memory cell **105** storing a first logic state (e.g., a logic 0) to threshold (e.g., permit a current flow, permit a current above a threshold current), and may not cause a memory cell **105** storing a second logic state (e.g., a logic 1) to threshold (e.g., may not permit a current flow, may permit a current below a threshold current).

[0047] In some cases, a memory array **200** may implement a multi-transistor architecture, such as a two-transistor architecture. For example, the memory array **200** may include pillars **220** coupled to an access line, such as a bit line **215**, using two transistors (e.g., the transistors **225-a** and **225-b**) positioned between the pillar **220** and the bit line **215**. As part of an access operation for a memory cell **105** coupled with the pillar **220** (e.g., the selected pillar), the memory array **200** may be configured to bias the bit line **215** to a first voltage and activate the two transistors **225-a-1** and **225-b-1** using a second voltage to couple the pillar **220** with the biased bit line **215**. Additionally, the memory array **200** may include circuitry, such as a set of pull up transistors and pull down transistors, to bias a gate of a first transistor **225-a-2** and a gate of a second transistor **225-b-2** coupling a second pillar to the bit line **215** (e.g., an unselected pillar) to a third and fourth voltage, respectively, which may deactivate at least one of the first or second transistor during the access operation. Accordingly, the unselected pillar may remain isolated from the bit line **215** during the access operation, and the transistors coupling the unselected pillar to the bit line **215** may undergo less electrical stress. Thus, by implementing the multi-transistor architecture, such as the two-transistor architecture, a memory array **200** may be implemented with reduced leakage through deactivated transistors, which may support reduced power consumption, increased accuracy of access operations, or any combination thereof, among other benefits.

[0048] FIG. 4 illustrates an example of a memory array **400** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array **400** may include aspects of the memory array **200** described with reference to FIGS. 2, 3A, and 3B. For example, the memory array **400** may include an arrangement of conductive pillars, such as a conductive pillar **405-a** and a conductive pillar **405-b**, which may be examples of pillars **220** described with reference to FIGS. 2, 3A, and 3B. The memory array **400** may also include various arrangements of word lines **205** (e.g.,

arranged in a comb structure) and memory cells **105** (e.g., a three-dimensional array of memory cells **105**), which also may be examples of the respective components described with reference to FIGS. 2, 3A, and 3B, though such components are omitted from FIG. 4 for the sake of illustrative clarity.

[0049] In some examples, the conductive pillars **405-a** and **405-b** may extend through one or more levels of memory cells **105** (e.g., levels **230**) and may be coupled with one or more memory cells **105** (e.g., two memory cells **105**) at each level **230**. For example, at each level, one or more memory cells **105** may be coupled with (e.g., coupled physically between, coupled electrically between, or both) a conductive pillar and a respective word line **205**. In some examples, the conductive pillars **405-a** and **405-b** may couple respective memory cells with a bit line **420**, which may be an example of a bit line **215**. Although the illustrative example of memory array **400** includes circuitry associated with two conductive pillars, elements of the memory array **400** in accordance with examples as disclosed herein may be repeated for any quantity of conductive pillars.

[0050] The memory array **400** illustrates an example of a multi-transistor architecture (e.g., a two-transistor architecture), which may facilitate biasing the conductive pillars **405-a** and **405-b** (e.g., as part of an access operation) using a set of transistor to selectively couple a conductive pillar with the bit line **420**. For example, the conductive pillar **405-a** may be coupled with the bit line **420** through a first transistor **410-a** and a second transistor **415-a**. Additionally, the conductive pillar **405-b** may be coupled with the bit line **420** through a third transistor **410-b** and a fourth transistor **415-b**.

[0051] In some cases, the transistors **410-a**, **410-b**, **415-a**, and **415-b** may be examples of thin-film transistors (TFTs), such as metal-oxide silicon field effect (MOSFET) transistors and may include a cascode transistor or cascode configuration of transistors. A cascode may be an example of a two-stage amplifier that includes of a common-emitter stage transistor feeding into a common-base stage transistor. In some cases, a cascode may be represented in figures as a single transistor. For example, a terminal of the first transistor **410-a** may be coupled with a terminal of the second transistor **415-a** (e.g., in a common-emitter stage feeding into a common-base stage configuration). Additionally, the transistors **410-a**, **410-b**, **415-a**, and **415-b** may be n-type transistors (e.g., n-type MOSFET or n-mos transistors). Accordingly, the first transistor **410-a** and the second transistor **415-a** may electrically couple the conductive pillar **405-a** with the bit line **420** if a voltage difference between the bit line **420** and respective voltages of gates of the first transistor **410-a** and second transistor **415-a** exceed respective thresholds (e.g., the threshold voltages of the first transistor **410-a** and the second transistor **415-a**), and the third transistor **410-b** and the fourth transistor **415-b** may electrically couple the conductive pillar **405-b** with the bit line **420** if a voltage difference between the bit line **420** and respective voltages of gates of the third transistor **410-b** and fourth transistor **415-b** exceed respective thresholds (e.g., the threshold voltages of the third transistor **410-b** and the fourth transistor **415-b**).

[0052] The transistors **410-a**, **415-a**, **410-b**, and **415-b** may be coupled with a respective gate line. For example, a gate of the first transistor **410-a** may be coupled with a first gate line **425-a**, a gate of the second transistor **415-a** may be

coupled with a second gate line **430-a**, a gate of the third transistor **410-b** may be coupled with a third gate line **425-b**, and a gate of the fourth transistor **415-b** may be coupled with a fourth gate line **430-b**. The gate lines may be configured to independently bias respective gates of the transistors to a variety of voltages as part of access operations for memory cells coupled with the conductive pillars **405-a** and **405-b**. In some cases, the memory array **400** may include one or more drivers or decoders which may bias the gate lines **425-a**, **430-a**, **425-b**, and **430-b** to respective voltages as part of an access operation.

[0053] For example, a first access operation to store a first value in a memory cell coupled with the conductive pillar **405-a** (e.g., a selected pillar) may include biasing the bit line **420** to a positive first voltage and coupling the bit line **420** with the conductive pillar **405-a** (e.g., the access operation may be an example of a “select positive” operation). In such examples, the gate lines **425-a** and **430-b** may be biased to a second voltage higher than the first voltage, such that the difference between the second voltage and the first voltage may be equal to or exceed the threshold voltages of the first transistor **410-a** and the second transistor **415-a**. In an illustrative, non-limiting example, the bit line **420** may be biased to a voltage of 3.6 volts (V), while the first gate line **425-a** and the second gate line **430-a** may be biased to a voltage of 5V or to a supply voltage, though one skilled in the art may appreciate that other combinations of voltages are possible.

[0054] Additionally or alternatively, a second access operation to store a second value in a memory cell coupled with the conductive pillar **405-a** (e.g., the selected pillar) may include biasing the bit line **420** to a negative third voltage and coupling the bit line **420** with the conductive pillar **405-a** (e.g., the access operation may be an example of a “select negative” operation). In such examples, the gate lines **425-a** and **430-b** may be biased to a fourth voltage higher than the third voltage, such that the difference between the fourth voltage and the third voltage may be equal to or exceed the threshold voltages of the first transistor **410-a** and the second transistor **415-a**. In an illustrative, non-limiting example, the bit line **420** may be biased to a voltage of negative 3.6V, while the first gate line **425-a** and the second gate line **430-a** may be biased to a voltage of 0V, though one skilled in the art may appreciate that other combinations of voltages are possible.

[0055] As part of the first access operation, the second access operation, or both, the conductive pillar **405-b** (e.g., the unselected pillar) be isolated or decoupled from the bit line **420**. For example, the gate line **425-b** may be biased to fifth voltage and the gate line **430-b** may be biased to a sixth voltage less than the fifth voltage. In an illustrative, non-limiting example, the fifth voltage may be negative 3.6V, while the sixth voltage may be 0V (e.g., the fifth voltage may correspond to the third voltage and the sixth voltage may correspond to the fourth voltage) though one skilled in the art may appreciate that other combinations of voltages are possible. Accordingly, at least one of the third transistor **410-b** or the fourth transistor **415-b** (or both) may at least partially isolate or reduce conductance between the bit line **420** and the conductive pillar **405-b**. Thus, the transistors **410-b** and **415-b** may reduce leakage current from the conductive pillar **405-b** resulting an access operation for the conductive pillar **405-a**.

[0056] In some examples, while outside an access operation for the conductive pillar **405-a**, the conductive pillar **405-b**, or both (e.g., during an idle mode), the first gate line **425-a** and the third gate line **425-b** may respectively bias the first transistor **410-a** and the third transistor **410-b** to the fifth voltage, while the second gate line **430-a** and the fourth gate line **430-b** may respectively bias the second transistor **415-a** and the fourth transistor **415-b** to the sixth voltage. Accordingly, while outside the access operation, the transistors **410-a**, **410-b**, **415-a**, and **415-b** may isolate or decouple the conductive pillars **405-a** and **405-b** from the bit line **420**, which may further reduce leakage current associated with operations of the memory array **400**.

[0057] In some examples, the memory array **400** may include or may be coupled with one or more decoders or drivers used to bias the gate lines **425-a**, **425-b**, **430-a**, and **430-b**. For example, the first gate line **425-a** and the second gate line **430-a** may be coupled with a first decoder, while the third gate line **425-b** and the fourth gate line **430-b** may be coupled with a second decoder. The one or more decoders may be positioned at an end of a tile of the memory array **400**, between tiles of the memory array **400**, within a tile of the memory array **400**, or any combination thereof, as described in greater detail with reference to FIGS. 5A through 8.

[0058] FIG. 5A illustrates an example of a circuit **500-a** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The circuit **500-a** may include aspects of the memory array **400** described with reference to FIG. 4. For example, the circuit **500-a** may include or may be coupled with a first gate line **525** and a second gate line **530**, which may respectively be examples of the first gate line **425** and the second gate line **430** as described with reference to FIG. 4. The circuit **500-a** may be an example of a circuit used to bias the gate lines **425** and **430** to various voltages.

[0059] The circuit **500-a** may be coupled with or may be an example of a decoder configured to bias the first gate line **525** and the second gate line **530** to one or more voltages as part of an access operation. For example, the circuit **500-a** may bias the first gate line **525** and the second gate line **530** to a first voltage as part of a first access operation (e.g., to 5V, as part of a “select positive” operation as described with reference to FIG. 4). Additionally, the circuit **500-a** may bias the first gate line **525** and the second gate line **530** to a second voltage less than the first voltage as part of a second access operation (e.g., to 0V, as part of a “select negative” operation as described with reference to FIG. 4). In some examples, the circuit **500-a** may receive one or more input signals at one or more components of the circuit **500-a**, which may determine or generate the voltages used for the access operations.

[0060] For example, the circuit **500-a** may include a first transistor **505**. The first transistor **505** may include a first terminal coupled with a first signal node **550** and a second terminal coupled with the second gate line **530**. Additionally, the first transistor **505** may include a gate coupled with a second signal node **540-a**. In some cases, the first transistor **505** may be an example of a p-type MOSFET (p-mos) transistor, and accordingly, the first transistor **505** may couple the first signal node **550** with the second gate line **530** if the difference between a voltage at the gate of the first transistor **505** and a voltage at a terminal of the first transistor **505** is less than or equal to a threshold of the first

transistor **505** (e.g., a voltage threshold). The circuit **500-a** may additionally include a second transistor **510**. The second transistor **510** may include a first terminal coupled with a supply node **555** and a gate coupled with a third signal node **545**. In some cases, the first transistor **505** may be an example of a p-mos transistor, and the second transistor **510** may be an example of an n-mos transistor.

[0061] The circuit **500-a** may include one or more transistors positioned between the first transistor **505** and the second transistor **510**. For example, the circuit **500-a** may include a third transistor **520** and a fourth transistor **535**. The third transistor **520** may be an example of a p-mos transistor, and may include a first terminal coupled with a terminal of the first transistor **505** and the second gate line **530**, such that the second gate line **530** may be positioned between the first transistor **505** and the third transistor **520**. Additionally the third transistor **520** may include a second terminal coupled with the first gate line **525** and a first terminal of the third transistor **535**, such that the first gate line **525** may be positioned between the third transistor **520** and the fourth transistor **535**. In some cases, the fourth transistor **535** may be an example of an n-mos transistor, and may include a second terminal coupled with a terminal of the second transistor **510**. Accordingly, the first transistor **505**, the second transistor **510**, the third transistor **520**, and the fourth transistor **535** may be connected in series. In some examples, the third transistor **520** and the fourth transistor **535** may each include a gate coupled with a ground node or a virtual ground node.

[0062] In some cases, the circuit **500-a** may include a fifth transistor **515** coupled with the second gate line **530**. For example, the fifth transistor may be an example of an n-mos transistor and may include a first terminal coupled with the second gate line **530** and a second terminal coupled with a ground node of a virtual ground node. Additionally, the fifth transistor **515** may include a gate coupled with a third signal node **540-b**. In some cases, the third signal node **540-b** may carry or be biased with a same signal as the second signal node **540-a**. Additionally or alternatively, the second signal node **540-a** may be coupled with the third signal node **540-b**, or may be the same node as the third signal node **540-b**, such that the gate of the first transistor **505** and the gate of the fifth transistor **515** receive the same or similar bias voltages.

[0063] The circuit **500-a** may be configured to bias the first gate line **525** and the second gate line **530** to one or more voltages as part of an access operation. For example, as part of a first access operation to store a first value in a memory cell coupled with a pillar coupled with the first gate line **525** and the second gate line **530** (e.g., a selected pillar), the second signal node **540-a** and the third signal node **540-b** may apply a low signal (e.g., a logic “0”, a low voltage) to the gate of the first transistor **505** and the gate of the fifth transistor **515**. Accordingly, the low signal may activate the first transistor **505** and may deactivate the fifth transistor **515**. Thus, the first gate line **525** and the second gate line **530** may couple with the first signal node **550**. Additionally, because the fifth transistor **515** may be deactivated during the first access operation, the second gate line **530** may be isolated from the virtual ground node.

[0064] In some cases, as part of the first access operation, the first signal node **550** may be biased to a first voltage, which may bias the first gate line **525** and the second gate line **530** to the first voltage. Additionally or alternatively, as part of the first access operation, a bit line may be biased to

a second voltage less than the first voltage (e.g., as part of the “select positive” operation as described with reference to FIG. 4). In an illustrative, non-limiting example, the bit line may be biased to a voltage of 3.6 volts (V), while the first gate line **525** and the second gate line **530** may be biased to a voltage of 5V or to a supply voltage, though one skilled in the art may appreciate that other combinations of voltages are possible.

[0065] Additionally or alternatively, as part of a second access operation to store a second value in a memory cell coupled with a pillar coupled with the first gate line **525** and the second gate line **530** (e.g., a selected pillar), the second signal node **540-a** and the third signal node **540-b** may apply a high signal (e.g., a logic “1”, a high voltage) to the gate of the first transistor **505** and the gate of the fifth transistor **515**. Accordingly, the high signal may deactivate the first transistor **505** and may activate the fifth transistor **515**. Thus, the first gate line **525** and the second gate line **530** may couple with the virtual ground node (e.g., through the fifth transistor **515**), which may bias the first gate line **525** and the second gate line **530** to the voltage of the virtual ground node (e.g., to 0V).

[0066] As part of the second access operation, the bit line may be biased to a negative voltage less than the voltage of the virtual ground node (e.g., as part of the “select negative” operation as described with reference to FIG. 4). In an illustrative, non-limiting example, the bit line may be biased to a voltage of negative 3.6 volts (V), while the first gate line **525** and the second gate line **530** may be biased to a voltage of 0V, though one skilled in the art may appreciate that other combinations of voltages are possible.

[0067] In some examples, while outside an access operation (e.g., while in an idle mode), the circuit **500-a** may bias the first gate line **525** to a third voltage and may bias the second gate line **530** to a fourth voltage. For example, the third signal node **545** may activate the second transistor **510**, which may pull up a voltage of the supply node **555** and accordingly bias the first gate line **525** to the third voltage. Additionally, the second signal node **540-a** and the fourth signal node **540-b** may respectively apply a high signal to the gate of the first transistor **505** and the gate of the fifth transistor **515**, which may bias the second gate line **530** to the fourth voltage. In an illustrative, non-limiting example of the idle mode, the circuit **500-a** may bias the first gate line **525** may to a voltage of negative 3.6V, and may bias the second gate line **530** may to a voltage of 0V. Further, during the idle mode, the bit line may be biased to a voltage of 0V, though one skilled in the art may appreciate that other combinations of voltages are possible.

[0068] In some cases, the circuit **500-a** may include one or more inverters to bias the first gate line **525** and the second gate line **530**. For example, the fifth transistor **515** may be replaced with an inverter coupled with the second gate line **530**. In such examples, the second gate line **530** may not be coupled with (e.g., may be isolated from) the first transistor **505**, the second transistor **510**, the third transistor **520**, and the fourth transistor **535**. Additionally, an input of the inverter may be coupled with the third signal node **540-b**, while an output of the inverter may be coupled with the second gate line **530**. Accordingly, if the third signal node **540-b** carries a high signal, then the inverter may bias the second gate line **530** to a low signal or low voltage. Alternatively, if the third signal node **540-b** carries a low

signal, then the inverter may bias the second gate line 530 to a high signal or high voltage.

[0069] Additionally or alternatively, the fifth transistor 515 may be replaced with a pair of inverters coupled with the first gate line 525 and the second gate line 530. In such examples, the first gate line 525 may be coupled with an input of a first inverter of the pair of the inverters. An output of the first inverter may be coupled with an input of a second inverter of the pair of inverter, while an output of the second inverter may be coupled with the second gate line 530. Accordingly, if the first gate line 525 is biased to a high voltage (e.g., using the first transistor 505, the second transistor 510, the third transistor 520, and the fourth transistor 535), the second gate line 530 may be biased to a high voltage. Alternatively, if the first gate line 525 is biased to a low voltage, the second gate line 530 may be biased to a low voltage.

[0070] FIG. 5B illustrates an example of a memory array 500-b that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array 500-b may include aspects of the memory array 400, the circuit 500-a or both. For example, the memory array 500-b may include a circuit 560-a, which may be an example of the circuit 500-a. Additionally, the memory array 500-b may include one or more vias, such as the via 565-a, selectively coupled with a first gate line 580-a and a second gate line 585-a. The via 565-a may be coupled with the first gate line 580-a and the second gate line 585-a using a first transistor 570-a and a second transistor 575-a. Additionally, the via 565-a may be coupled with a bit line 573-a. In some cases, the via 565-a, the bit line 573-a, the first transistor 570-a, the second transistor 575-a, the first gate line 580-a, the second gate line 585-a, or any combination thereof may be examples of the corresponding components as described with reference to FIGS. 3A, 3B, and 4. For example, the first transistor 570-a may be an example of the transistors 225-a and 410 and the second transistor 575-a may be an example of the transistors 225-b and 415 described with reference to FIGS. 3A, 3B, and 4.

[0071] The memory array 500-b may illustrate an example of positioning the circuit 560-a between a first set of vias 590-a and a second set of vias 590-b. In some examples, the first set of vias 590-a may be part of a first tile of the memory array 500-b, while the second set of vias 590-b may be part of a second tile of the memory array 500-b. Accordingly, the circuit 560-a may be positioned between tiles of the memory array 500-b. Additionally or alternatively, the first set of vias 590-a may be part of a first set of interleaving comb structures (e.g., a first set of word lines 205) of a first tile of the memory array 500-b, while the second set of vias 590-b may be part of a second set of interleaving comb structures of the first tile of the memory array 500-b. Accordingly, the circuit 560-a may be positioned between sets of interleaving comb structures of the memory array 500-b.

[0072] In some cases, the circuit 560-a may be coupled with the first gate line 580-a and the second gate line 585-a using respective vias 583-a and 587-a. The vias 583-a and 587-a may not be coupled with memory cells of the memory array 500-b, and may respectively couple directly with the first gate line 580-a and the second gate line 585-a (e.g., the vias 583-a and 587-a may be examples of placeholder conductive pillars or vias). In some cases, the vias 583-a and 587-a may be orthogonal to the first gate line 580-a and the

second gate line 585-a. In some cases, the vias 583-a and 587-a may form or may be part of a wall of vias which may increase the complexity of coupling the circuit 560-a with the first gate line 580-a and the second gate line 585-a. For example, the wall of vias may interrupt the second gate line 585-a, as illustrated in FIG. 5B. Accordingly, the memory array 500-b may include additional circuitry to bias the second gate line 585-a (e.g., the portion of the second gate line 585-a coupled with the second set of vias 590-b), as described with greater detail with reference to FIGS. 7A, 7B, 7C, and 8.

[0073] FIG. 5C illustrates an example of a memory array 500-c that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array 500-c may include aspects of the memory array 400, the circuit 500-a, the memory array 500-b, or any combination thereof. For example, the memory array 500-c may include a circuit 560-b, which may be an example of the circuit 500-a. Additionally, the memory array 500-c may include one or more vias, such as the via 565-b, selectively coupled with a first gate line 580-b and a second gate line 585-b. The via 565-b may be coupled with the first gate line 580-b and the second gate line 585-b using a first transistor 570-b and a second transistor 575-b. Additionally, the via 565-a may be coupled with a bit line 573-a. In some cases, the via 565-b, the bit line 573-b, the first transistor 570-b, the second transistor 575-b, the first gate line 580-b, the second gate line 585-b, or any combination thereof may be examples of the corresponding components as described with reference to FIGS. 3A, 3B, and 4. For example, the first transistor 570-b may be an example of the transistors 225-a and 410 and the second transistor 575-b may be an example of the transistors 225-b and 415 described with reference to FIGS. 3A, 3B, and 4.

[0074] The memory array 500-c may illustrate an example of positioning the circuit 560-b at an end a set of vias 590-c. In some examples, the set of vias 590-c may be part of a tile of the memory array 500-c, such that the circuit 560-b may be positioned at an end of one or more tiles of the memory array 500-c. In some cases, the circuit 560-b may be coupled with the first gate line 580-b and the second gate line 585-b using respective vias 583-b and 587-b. The vias 583-a and 587-a may not be coupled with memory cells of the memory array 500-b (e.g., the vias 583-b and 587-b may be examples of placeholder conductive pillars or vias). In some cases, the vias 583-b and 587-b may be orthogonal to the first gate line 580-b and the second gate line 585-b., and may respectively couple directly with the first gate line 580-a and the second gate line 585-a.

[0075] In some cases, the memory array 500-c may include a circuit 563. The circuit 563 may be an example of the fifth transistor 515 as described with reference to FIG. 5A. For example, the circuit 563 may include a TFT coupled with a virtual ground node and the second via 587-b, as illustrated in FIG. 5C. In some cases, a gate of the TFT transistor may be coupled with the second signal node 540-a, as described with reference to FIG. 5A.

[0076] FIG. 6A illustrates an example of a circuit 600-a that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The circuit 600 may be coupled with or may be an example of a decoder configured to bias a first gate line and the second gate line, such as the first gate line

425 and the second gate line **430** as described with reference to FIG. 4, to one or more voltages as part of an access operation. For example, the circuit **600** may bias the second gate line to a first voltage as part of a first access operation (e.g., to 5V, as part of a “select positive” operation as described with reference to FIG. 4) using the first gate line. Additionally, the circuit **600** may bias the second gate line to a second voltage less than the first voltage as part of a second access operation (e.g., to 0V, as part of a “select negative” operation as described with reference to FIG. 4) using the first gate line. In some examples, the circuit **600** may receive one or more input signals at one or more components of the circuit **600**, which may determine or generate the voltages used for the access operations.

[0077] The circuit **600** may include a pair of inverters, such as a first inverter **605** and a second inverter **610**, coupled with the first gate line and the second gate line. In such examples, the first gate line may be coupled with an input **615** of the first inverter **605** of the pair of the inverters. An output of the first inverter **605** may be coupled with an input of the second inverter **610**, while an output **620** of the second inverter **610** may be coupled with the second gate line. Accordingly, the circuit **600** may bias the second gate line to a first voltage if the first gate line is biased to a high voltage (e.g., using the first transistor **505**, the second transistor **510**, the third transistor **520**, and the fourth transistor **535**). Alternatively, the circuit **600** may bias the second gate line to a second voltage if the first gate line is biased to a low voltage, as described in greater detail with reference to FIG. 6C.

[0078] In some cases, the inverters **605** and **610** may each include a set of supply nodes. For example, the inverter **605** may include a supply node **625-a** and a supply node **630-a**, while the second inverter **610** may include a supply node **625-b** and a supply node **630-b**. The voltage supplied to the supply nodes **625-a** and **625-b** (e.g., the high supply nodes) may correspond to the first voltage, while the voltage supplied to the supply nodes **630-a** and **630-b** (e.g., the low supply nodes) may correspond to the second voltage. Accordingly, the circuit **600** may output the first voltage if the input voltage (e.g., the voltage of the first gate line) is less than or equal to a threshold (e.g., less than or equal to the voltage supplied to the supply nodes **630-a** and **630-b**). Additionally or alternatively, the circuit **600** may output the second voltage if the input voltage exceeds the threshold.

[0079] FIG. 6B illustrates an example of a memory array **601** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array **601** may include aspects of the memory array **400**, the memory array **500-b**, the memory array **500-c**, the circuit **600**, or any combination thereof. For example, the memory array **601** may include a circuit **635**, which may be an example of the circuit **600**. Additionally, the memory array **601** may include one or more sets of vias, such as the set of vias **640-a** and **640-b**, selectively coupled with a first gate line **645** and a second gate line **650**. In some cases, the set of vias **640-a** and **640-b**, the first gate line **645**, the second gate line **650**, or any combination thereof may be examples of the corresponding components as described with reference to FIGS. 4, 5A, and 5B.

[0080] The memory array **601** may illustrate an example of positioning the circuit **635** between the first set of vias **640-a** and the second set of vias **640-b**. In some examples,

the first set of vias **640-a** may be part of a first tile of the memory array **601**, while the second set of vias **640-b** may be part of a second tile of the memory array **601**. Accordingly, the circuit **635** may be positioned between tiles of the memory array **601**. Additionally or alternatively, the first set of vias **640-a** may be part of a first set of interleaving comb structures (e.g., a first set of word lines **205**) of a first tile of the memory array **601**, while the second set of vias **640-b** may be part of a second set of interleaving comb structures of the first tile of the memory array **601**. Accordingly, the circuit **635** may be positioned between sets of interleaving comb structures of the memory array **601**.

[0081] In some cases, the circuit **635** may be coupled with the first gate line **645** using a via **653**. The via **653** may not be coupled with memory cells of the memory array **601**, and may couple with the first gate line **645**. In some cases, the via **653** may form or may be part of a wall of vias which may increase the complexity of coupling the circuit **635** with the first gate line **645** and the second gate line **650**. For example, the wall of vias may interrupt the second gate line **650**, as illustrated in FIG. 5B, which may split the second gate line **650** into a first portion **650-a** and a second portion **650-b**.

[0082] Accordingly, the memory array **601** may include additional circuitry to bias the second gate line **650** (e.g., the first portion **650-a** and the second portion **650-b**). For example, the circuit **635** may include a second set of inverters (e.g., a third inverter and a fourth inverter), in addition to the first inverter **605** and the second inverter **610**. In some examples, the first inverter **605** and the second inverter **610** may be configured to bias the first portion of the second gate line **650-a**, while the third inverter and the fourth inverter may be configured to bias the second portion of the gate line **650-b**. In some examples, the first portion of the gate line **650-a** and the second portion of the gate line **650-b** may be coupled with the circuit **635** using respective vias, as illustrated in FIG. 6B.

[0083] In some example, the first gate line **645** and the second gate line **650** may be biased using the same signal. For example, the gate line **645** may be biased by a circuit (not shown). The voltage on the gate line **645** may be conducted along via **653** to the circuit **635**. The circuit **635** may include an input signal node, which may receive an input signal from the gate line **645** (e.g., using the via **653**). Because the sets of inverters of the circuit **635** may be coupled with the input signal node, the input signal may route through the sets of inverters to bias the second gate line **650**. Accordingly, the memory array **601** may be configured to bias the first gate line **645** and the second gate line **650** using a single signal.

[0084] In some example, the circuit **635** may bias the first gate line **645** and the second gate line **650** using the same signal. For example, the circuit **635** may include an input signal node, which may apply an input signal to the gate line **645** (e.g., using the via **653**). Because the sets of inverters of the circuit **635** may be coupled with the input signal node, the input signal may route through the sets of inverters to bias the second gate line **650**.

[0085] FIG. 6C illustrates an example of a timing diagram **602** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The timing diagram **602** may illustrate the voltage **655** of components of the memory array **601** over time **660** during an access operation. For example, the timing diagram **602** may illustrate a voltage of a first gate

line 665, a voltage of a second gate line 670, and a voltage of a bit line 675. In some cases, the first gate line, the second gate line, and the bit line may be examples of the corresponding components as described with reference to FIGS. 4 through 6B.

[0086] The timing diagram 602 may illustrate an example of a select positive operation (e.g., as described with reference to FIG. 4) for a memory array, such as the memory array 500-b, the memory array 500-c, or the memory array 601. Initially, the memory array may be in an idle mode. For example, a signal node may apply a first voltage 685 to the first gate line (e.g., a negative voltage, such as negative 3.6V). Because the second gate line may be coupled with the first gate line using a pair of inverters (e.g., as described with reference to FIGS. 6A and 6B), the first voltage may pass through the pair of inverters to bias the second gate line to a second voltage 690. (e.g., 0V).

[0087] As part of the select positive operation, the signal node may apply a third voltage 695 (e.g., a voltage higher than the second voltage 690, such as 5V) to the first gate line. Accordingly, the third voltage 695 may pass through the pair of inverters to bias the second gate line to the third voltage 695. In some cases, the part of inverters may act as a buffer or delay component. Thus, there may be a delay 680-a between the voltage of the first gate line 665 being biased to the third voltage 695 and the voltage of the second gate line 670 being biased to the third voltage 695.

[0088] In some cases, while the voltage of the first gate line 665 and the voltage of the second gate line 670 are biased to the third voltage 695, the voltage of the bit line 675 may be biased to a fourth voltage 697. In some cases, the fourth voltage 697 may be less than the third voltage 695 (e.g., the fourth voltage 697 may be 3.6V). Accordingly, the first gate line and the second gate line may activate respective transistors coupled with a via to couple the via with the bit line to access a memory cell (e.g., as described with reference to FIG. 4). In some cases, after accessing the memory cell, the signal node may remove the third voltage 695 from the first gate line and may apply the first voltage 685 to the first gate line. Accordingly, the voltage of the first gate line 665 may fall to the first voltage 685, and the voltage of the second gate line 670 may fall to the second voltage 690. Additionally, the voltage of the bit line 675 may return to the second voltage 690. In some cases, because the pair of inverters may act as a buffer, there may be a second delay 680-b between the fall of the voltage of the first gate line 665 and the fall of the voltage of the second gate line 670.

[0089] In some cases, to perform a select negative operation, the signal node may apply the second voltage 690 to the first gate line (e.g., instead of the third voltage 695, as in the select positive operation). Accordingly, the voltage of the first gate line 665 may rise to the second voltage 690, while the voltage of the second gate line 670 may remain at the second voltage 690. In some cases, while the first gate line is biased to the second voltage 690, the bit line may be biased to a fifth voltage. The fifth voltage may be lower than the second voltage 690 (e.g., the fifth voltage may be approximately negative 3.6V). Accordingly, the first gate line and the second gate line may activate respective transistors coupled with a via to couple the via with the bit line to access a memory cell (e.g., as described with reference to FIG. 4).

[0090] FIG. 7A illustrates an example of a memory array 700-a that supports shared decoder architecture for three-

dimensional memory arrays in accordance with examples as disclosed herein. The memory array 700-a may include a first tile 705-a and a second tile 710-a. The first tile 705-a and the second tile 710-a may each include aspects and components of the memory array 400, as described with reference to FIG. 4. For example, the first tile 705-a and the second tile 710-a may include one or more vias extending through a set of levels of the memory array 700-a, the vias selectively coupling respective memory cells to respective bit lines. In some cases, each via may be coupled with a respective pillar using a first transistor and a second transistor, where a gate of the first transistor is coupled with a first gate line, such as the first gate line 715-a, and a gate of the second transistor is coupled with a second gate line, such as the second gate line 720-a.

[0091] The memory array 700-a may include one or more circuits to bias the first gate line 715-a and the second gate line 720-a as part of an access operation, such as the circuit 725-a, positioned between the first tile 705-a and the second tile 710-a. For example, the circuit 725-a may be positioned at an edge of the first tile 705-a adjoining an edge of the second tile 710-a. The circuit 725-a may include aspects of or may be an example of the circuit 500-a, the circuit 600, or a combination thereof. In some cases, positioning the circuit 725-a between tiles may allow for increased speed of an access operation compared to other configurations, for example by mitigating the distance between an accessed memory cell and the circuit 725-a. However, positioning the circuit 725-a between the first tile 705-a and the second tile 710-a may increase the complexity of coupling the first gate line 715-a and the second gate line 720-a with the circuit 725-a. For example, each circuit 725-a of the one or more circuits, may include a via coupled with the first gate line 715-a, for example as described with reference to FIGS. 5B, 5C, and 6B. In such examples, the vias may form a wall of vias 730-a, which may interrupt the second gate line 720-a (e.g., may split the second gate line 720-a into two portions), as illustrated in FIG. 7A. Accordingly, the circuit 725-a may include additionally circuitry to bias both portions of the second gate line 720-a, such as a first set of inverters to bias a first portion of the second gate line 720-a and a second set of inverters to bias the second portion of the second gate line 720-a, as described with reference to FIG. 6B.

[0092] FIG. 7B illustrates an example of a memory array 700-b that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array 700-b may include a first tile 705-b and a second tile 710-b. The first tile 705-b and the second tile 710-b may each include aspects and components of the memory array 400, as described with reference to FIG. 4. For example, the first tile 705-b and the second tile 710-b may include one or more vias extending through a set of levels of the memory array 700-b, the vias selectively coupling respective memory cells to respective bit lines. In some cases, each via may be coupled with a respective pillar using a first transistor and a second transistor, where a gate of the first transistor is coupled with a first gate line, such as the first gate line 715-b, and a gate of the second transistor is coupled with a second gate line, such as the second gate line 720-b.

[0093] The memory array 700-b may include one or more circuits to bias the first gate line 715-b and the second gate line 720-b as part of an access operation, such as the circuit 725-b, positioned at an end of the first tile 705-b and the

second tile **710-b** (e.g., rather than between the first tile and the second tile, as illustrated in FIG. 7A). For example, the circuit **725-b** may be positioned at an edge of the first tile **705-b** that does not adjoin an edge of the second tile **710-b**. Additionally or alternatively, the circuit **725-b** may be positioned at a first edge of the first tile **705-b** opposite a second edge of the first tile **705-b** adjoining an edge of the second tile **710-b**. Additionally or alternatively The circuit **725-b** may include aspects of or may be an example of the circuit **500-a**, the circuit **600**, or a combination thereof. In some cases, positioning the circuit **725-b** at the end of the first tile **705-b** and the second tile **710-b** may allow the circuit **725-b** to couple with the first gate line **715-b** and the second gate line **720-b** without interrupting the second gate line **720-b** with the associated wall of vias **730-b**. In some examples, positioning the circuit **725-b** at the end of the first tile **705-b** and the second tile **710-b** may increase the length of the first gate line **715-b** and the second gate line **720-b**. However, the circuit **725-b** may be configured to bias the relatively longer first gate line **715-a** and second gate line **720-b** without a significant decrease in the speed or performance of an access operation.

[0094] FIG. 7C illustrates an example of a memory array **700-c** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory array **700-c** may include a first tile **705-c** and a second tile **710-c**. The first tile **705-c** and the second tile **710-c** may each include aspects and components of the memory array **400**, as described with reference to FIG. 4. For example, the first tile **705-c** and the second tile **710-c** may include one or more pillars extending through a set of levels of the memory array **700-c**, the pillars selectively coupling respective memory cells to respective bit lines. In some cases, each pillar may be coupled with a respective via using a first transistor and a second transistor, where a gate of the first transistor is coupled with a first gate line, such as the first gate line **715-c**, and a gate of the second transistor is coupled with a second gate line, such as the second gate line **720-c**.

[0095] The memory array **700-c** may include one or more circuits to bias the first gate line **715-c** and the second gate line **720-c** as part of an access operation, such as the circuit **725-c**, the circuit **735-c**, and the circuit **740-c**. In some examples, the circuit **725-c** may be coupled with the first gate line **715-c**, and may include one or more transistors to bias the first gate line **715-c** to a first voltage, such as the transistor **505**, the transistor **520**, or both as described with reference to FIG. 5A (e.g., the “pull up” transistors). The circuit **735-c** may be coupled with the first gate line **715-c**, and may include one or more transistors to bias the first gate line **715-c** to a second voltage, such as the transistor **510**, the transistor **535**, or both as described with reference to FIG. 5A (e.g., the “pull down” transistors). The circuit **740-c** may be coupled with the second gate line **720-c**, and may include circuitry to bias the second gate line, such as the transistor **515** or one or more inverters, as described with reference to FIG. 5A. In some cases, the circuit **725-c** and the circuit **740-c** may be positioned at a first end of the first tile **705-c** and the second tile **710-c**, while the circuit **735-c** may be positioned at a second end of the first tile **705-c** and the second tile **710-c** opposite the first end.

[0096] FIG. 7D illustrates an example of a memory array **700-d** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as

disclosed herein. The memory array **700-d** may include a first tile **705-d** and a second tile **710-d**. The first tile **705-d** and the second tile **710-d** may each include aspects and components of the memory array **400**, as described with reference to FIG. 4. For example, the first tile **705-d** and the second tile **710-d** may include one or more pillars extending through a set of levels of the memory array **700-d**, the pillars selectively coupling respective memory cells to respective bit lines. In some cases, each pillar may be coupled with a respective via using a first transistor and a second transistor, where a gate of the first transistor is coupled with a first gate line, such as the first gate line **715-d**, and a gate of the second transistor is coupled with a second gate line, such as the second gate line **720-d**.

[0097] Additionally, the first tile **705-d** and the second tile **710-d** may include one or more sets of interleaved comb structures, such as a first interleaved comb structure **745** and a second interleaved comb structure **750**, coupled with respective sets of pillars. Each interleaved comb structure may include a first set of word lines and a second set of word lines (e.g., an even word line and an odd word line). In some examples, the first tile **705-d** may include one or more placeholder vias (e.g., vias which are not coupled with a memory cell, as described with reference to FIGS. 5B, 5C, and 6B) between the first interleaved comb structure **745** and the second interleaved comb structure **750**. In such examples, the memory array **700-d** may include one or more circuits which utilize the placeholder vias, placeholder vias between the first tile **705-d** and **710-d**, or both to bias the first gate line **715-d** and the second gate line **720-d** as part of an access operation.

[0098] An interleaved comb structure for word lines (or other access lines) is shown and described with reference to the even word line **205-a-32** and the odd word line **205-a-31** of FIG. 2. The even word line **205-a-32** includes a first plurality of conductive lines extending into the array of memory cells and coupled with a plurality of memory cells. The first plurality of conductive lines are coupled with a common conductive line configured to drive the first plurality of conductive lines. The odd word line **205-a-31** includes a second plurality of conductive lines extending into the array of memory cells and coupled with the plurality of memory cells. The second plurality of conductive lines are coupled with a common conductive line configured to drive the second plurality of conductive lines. The first plurality of conductive lines are interleaved (e.g. interdigitated) with the second plurality of conductive lines such that every other word line in the memory device is an even word line **205-a-32** and the other word lines are the odd word line **205-a-31**.

[0099] The first interleaved comb structure **745** includes an even word line and its corresponding first plurality of conductive lines interleaved with a corresponding second plurality of conductive lines of an odd word line. A single even word line may be an example of a first plurality of conductive lines coupled with a single common conductive line and all conductive lines being biased by a single driver. Similarly, a single odd word line may be an example of a second plurality of conductive lines coupled with a single common conductive line and all conductive lines being biased by a single driver. A single interleaved comb structure may be an example a single even word line interleaved with a single odd word line. Thus, the first interleaved comb structure **745** may include a first even word line interleaved

with a first odd word line and the second interleaved comb structure **750** may include a second even word line interleaved with a second odd word line.

[0100] For example, the memory array **700-d** may include a circuit **725-d** coupled with the first gate line **715-d**. In some cases, the circuit **725-d** may include aspects of the circuit **500-a**, such as the transistor **505**, the transistor **510**, the transistor **520**, the transistor **535**, or any combination thereof. Accordingly, the circuit **725-d** may be configured to bias the first gate line **715-d** to one or more voltages as part of an access operation. In some cases, the circuit **725-d** may be positioned between the first tile **705-d** and **710-d**, as illustrated in FIG. 7D. Additionally or alternatively, the circuit **725-d** may be positioned between the first interleaved comb structure **745** and the second interleaved comb structure **750**.

[0101] The memory array **700-d** may further include a circuit **740-d** coupled with the second gate line **720-d**. In some cases, the circuit **740-d** may include aspects of the circuit **500-a**, such as the transistor **515** or a single inverter as described with reference to FIG. 5A.

[0102] Additionally or alternatively, the circuit **700-d** may include aspects of the circuit **600**. The circuit **725-d** may be used to bias the first gate line **715-d** using a circuit positioned between tiles. The position of the circuit **725-d** in-between the tiles (e.g., tiles **705-d** and **710-d**) may be used so that the vias that couple the first gate lines **715-d** with the circuits **725-d** without disrupting other aspects of the memory device. The circuit **740-d** may be an example of the circuit **601**, such as the first inverter **605** and the second inverter **610** as described with reference to FIGS. 6A and 6B. Accordingly, the circuit **740-d** may be coupled with the first gate line **715-d** using a via (e.g., via **653**). In response to the circuit **725-d** biasing the first gate line **715-d**, the circuit **740-d** may receive the signal and may drive a modified signal on the second gate line **720-d**. For example, the circuit **740-d** may use the inverters to modify the signal received over the first gate line **715-d** to bias the second gate line **720-d**. In some cases, the circuit **740-d** may be positioned between the first interleaved comb structure **745** and the second interleaved comb structure **750** associated with word lines of the memory device. Additionally or alternatively, the circuit **725-d** may be positioned between the first tile **705-d** and **710-d**.

[0103] FIG. 8 shows a block diagram **800** of a memory device **820** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The memory device **820** may be an example of aspects of a memory device as described with reference to FIGS. 1 through 7D. The memory device **820**, or various components thereof, may be an example of means for performing various aspects of shared decoder architecture for three-dimensional memory arrays as described herein. For example, the memory device **820** may include an access operation component **825**, a coupling component **830**, an access line biasing component **835**, a gate line biasing component **840**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

[0104] The access operation component **825** may be configured as or otherwise support a means for performing an access operation for a memory cell that is coupled with a word line and a pillar extending through a plurality of levels

of a memory array. The coupling component **830** may be configured as or otherwise support a means for coupling the pillar with a bit line by biasing a first gate line and a second gate line to a first voltage based at least in part on applying a first signal to a first transistor coupled with the first gate line and a first signal node and on applying a second signal to a second transistor coupled with a supply node and the first gate line. The access line biasing component **835** may be configured as or otherwise support a means for biasing the bit line to a second voltage based at least in part on coupling the pillar with the bit line.

[0105] In some examples, biasing the second gate line to the first voltage is further based at least in part on applying the first signal to a third transistor coupled with the second gate line and a virtual ground node.

[0106] In some examples, applying the first signal to the first transistor and the third transistor is based at least in part on applying, via a second signal node, the first signal to a gate of the first transistor and a gate of the third transistor.

[0107] In some examples, applying the second signal to the second transistor is based at least in part on applying, via a third signal node, the second signal to a gate of the second transistor.

[0108] In some examples, the access operation component **825** may be configured as or otherwise support a means for performing a second access operation for the memory cell. In some examples, the coupling component **830** may be configured as or otherwise support a means for coupling the pillar with the bit line by biasing the first gate line and the second gate line to a third voltage based at least in part on applying a third signal to the first transistor and applying a fourth signal to the second transistor, where the third voltage is less than the first voltage. In some examples, the access line biasing component **835** may be configured as or otherwise support a means for biasing the bit line to a fourth voltage based at least in part on biasing the first gate line and the second gate line to the third voltage, where the fourth voltage is less than the third voltage.

[0109] In some examples, the gate line biasing component **840** may be configured as or otherwise support a means for coupling the second gate line with the virtual ground node based at least in part on applying the third signal to the third transistor, where biasing the first gate line and the second gate line to the third voltage is based at least in part on applying the third signal.

[0110] In some examples, the gate line biasing component **840** may be configured as or otherwise support a means for coupling the first gate line with the first signal node based at least in part on applying the first signal to the first transistor, where biasing the first gate line and the second gate line to the first voltage as based at least in part on applying a third signal to the first signal node.

[0111] FIG. 9 shows a flowchart illustrating a method **900** that supports shared decoder architecture for three-dimensional memory arrays in accordance with examples as disclosed herein. The operations of method **900** may be implemented by a memory device or its components as described herein. For example, the operations of method **900** may be performed by a memory device as described with reference to FIGS. 1 through 8. In some examples, a memory device may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally or alternatively, the

memory device may perform aspects of the described functions using special-purpose hardware.

[0112] At **905**, the method may include performing an access operation for a memory cell that is coupled with a word line and a pillar extending through a plurality of levels of a memory array. The operations of **905** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **905** may be performed by an access operation component **825** as described with reference to FIG. 8.

[0113] At **910**, the method may include coupling the pillar with a bit line by biasing a first gate line and a second gate line to a first voltage based at least in part on applying a first signal to a first transistor coupled with the first gate line and a first signal node and on applying a second signal to a second transistor coupled with a supply node and the first gate line. The operations of **910** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **910** may be performed by a coupling component **830** as described with reference to FIG. 8.

[0114] At **915**, the method may include biasing the bit line to a second voltage based at least in part on coupling the pillar with the bit line. The operations of **915** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **915** may be performed by an access line biasing component **835** as described with reference to FIG. 8.

[0115] In some examples, an apparatus as described herein may perform a method or methods, such as the method **900**. The apparatus may include features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor), or any combination thereof for performing the following aspects of the present disclosure:

[0116] Aspect 1: A method, apparatus, or non-transitory computer-readable medium including operations, features, circuitry, logic, means, or instructions, or any combination thereof for performing an access operation for a memory cell that is coupled with a word line and a pillar extending through a plurality of levels of a memory array, the access operation including; coupling the pillar with a bit line by biasing a first gate line and a second gate line to a first voltage based at least in part on applying a first signal to a first transistor coupled with the first gate line and a first signal node and on applying a second signal to a second transistor coupled with a supply node and the first gate line; and biasing the bit line to a second voltage based at least in part on coupling the pillar with the bit line.

[0117] Aspect 2: The method, apparatus, or non-transitory computer-readable medium of aspect 1 where biasing the second gate line to the first voltage is further based at least in part on applying the first signal to a third transistor coupled with the second gate line and a virtual ground node.

[0118] Aspect 3: The method, apparatus, or non-transitory computer-readable medium of aspect 2 where applying the first signal to the first transistor and the third transistor is based at least in part on applying, via a second signal node, the first signal to a gate of the first transistor and a gate of the third transistor.

[0119] Aspect 4: The method, apparatus, or non-transitory computer-readable medium of aspect 3 where applying the

second signal to the second transistor is based at least in part on applying, via a third signal node, the second signal to a gate of the second transistor.

[0120] Aspect 5: The method, apparatus, or non-transitory computer-readable medium of any of aspects 2 through 4, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for performing a second access operation for the memory cell, the second access operation including; coupling the pillar with the bit line by biasing the first gate line and the second gate line to a third voltage based at least in part on applying a third signal to the first transistor and applying a fourth signal to the second transistor, where the third voltage is less than the first voltage; and biasing the bit line to a fourth voltage based at least in part on biasing the first gate line and the second gate line to the third voltage, where the fourth voltage is less than the third voltage.

[0121] Aspect 6: The method, apparatus, or non-transitory computer-readable medium of aspect 5, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for coupling the second gate line with the virtual ground node based at least in part on applying the third signal to the third transistor, where biasing the first gate line and the second gate line to the third voltage is based at least in part on applying the third signal.

[0122] Aspect 7: The method, apparatus, or non-transitory computer-readable medium of any of aspects 1 through 6, further including operations, features, circuitry, logic, means, or instructions, or any combination thereof for coupling the first gate line with the first signal node based at least in part on applying the first signal to the first transistor, where biasing the first gate line and the second gate line to the first voltage as based at least in part on applying a third signal to the first signal node.

[0123] It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.

[0124] An apparatus is described. The following provides an overview of aspects of the apparatus as described herein:

[0125] Aspect 8: An apparatus, including: a pillar extending through a plurality of levels of a memory array, where, at each level of the plurality of levels, one or more memory cells of the memory array are coupled with the pillar and a respective word line; a bit line; a first transistor and a second transistor in a series configuration with the first transistor, the first transistor and the second transistor configured to selectively couple the bit line with the pillar; a first gate line coupled with the first transistor; a second gate line coupled with the second transistor; a circuit coupled with the first gate line and the second gate line, the circuit including: a third transistor coupled with the first gate line and a first signal node; and a fourth transistor coupled with a supply node and the first gate line; and a decoder coupled with the circuit and configured to activate and deactivate the third transistor and the fourth transistor.

[0126] Aspect 9: The apparatus of aspect 8, where the circuit further includes: a fifth transistor positioned between the second gate line and a virtual ground node, where a gate of the third transistor and a gate of the fifth transistor are coupled with a second signal node.

[0127] Aspect 10: The apparatus of aspect 9, further including: a first tile including the pillar and a second tile including a second pillar extending through the plurality of levels of the memory array, the first tile adjoining the second tile at a first edge of the first tile, where locations of the third transistor, the fourth transistor, and the fifth transistor are associated with a second edge of the first tile opposite the first edge.

[0128] Aspect 11: The apparatus of any of aspects 9 through 10, further including: a first tile including the pillar and a second tile including a second pillar extending through the plurality of levels of the memory array, the first tile adjoining the second tile at a first edge of the first tile and a first edge of the second tile, where locations of the third transistor, the fourth transistor, and the fifth transistor are associated with the first edge of the first tile and the first edge of the second tile.

[0129] Aspect 12: The apparatus of any of aspects 9 through 11, further including: a first tile including the pillar and a second tile including a second pillar extending through the plurality of levels of the memory array, the first tile adjoining the second tile at a first edge of the first tile and a first edge of the second tile, where locations of the third transistor and the fifth transistor are associated with a second edge of the first tile opposite the first edge of the first tile and a location of the fourth transistor is associated with a second edge of the second tile opposite the first edge of the second tile.

[0130] Aspect 13: The apparatus of any of aspects 9 through 12, where the second gate line is coupled with the third transistor and the first gate line.

[0131] Aspect 14: The apparatus of any of aspects 9 through 13, where the fourth transistor and the fifth transistor are n-mos transistors and the third transistor is a p-mos transistor.

[0132] Aspect 15: The apparatus of any of aspects 9 through 14, where a gate of the third transistor and a gate of the fifth transistor are coupled with a third signal node and a gate of the fourth transistor is coupled with a fourth signal node.

[0133] Aspect 16: The apparatus of any of aspects 8 through 15, where: the first gate line includes a first conductive line coupled with a plurality of first transistors and a first via that extends orthogonal to the first conductive line, the first via coupling the first conductive line with the circuit; and the second gate line includes a second conductive line coupled with a plurality of second transistors and a second via that extends orthogonal to the first conductive line, the second via coupling the second conductive line with the circuit.

[0134] Aspect 17: The apparatus of any of aspects 8 through 16, where the circuit further includes: an inverter coupled with the second gate line, where a gate of the third transistor and an input of the inverter are coupled with a second signal node.

[0135] Aspect 18: The apparatus of any of aspects 8 through 17, where the circuit further includes: a fifth transistor coupled with the first gate line and the second gate line; and a sixth transistor coupled with the fourth transistor and the first gate line, where a gate of the fifth transistor and a gate of the sixth transistor are coupled with a virtual ground node.

[0136] Aspect 19: The apparatus of aspect 18, where the fifth transistor is a p-mos transistor and the sixth transistor is an n-mos transistor.

[0137] Aspect 20: The apparatus of any of aspects 8 through 19, where the one or more memory cells include a chalcogenide material.

[0138] An apparatus is described. The following provides an overview of aspects of the apparatus as described herein:

[0139] Aspect 21: An apparatus, including: a pillar extending through a plurality of levels of a memory array, where, at each level of the plurality of levels, one or more memory cells of the memory array are coupled with the pillar and a respective word line; a bit line; a first transistor and a second transistor in a series configuration with the first transistor, the first transistor and the second transistor configured to selectively couple the bit line with the pillar; a first gate line coupled with the first transistor; a second gate line coupled with the second transistor; a circuit coupled with the first gate line and the second gate line, the circuit including: a signal node coupled with the first gate line; a first inverter coupled with the signal node and the second gate line; and a second inverter coupled with the first inverter and the second gate line; and a decoder coupled with the circuit and configured to drive a signal to the signal node.

[0140] Aspect 22: The apparatus of aspect 21, further including: a first tile including the pillar and a second tile including a second pillar extending through the plurality of levels of the memory array, the first tile adjoining the second tile at a first edge of the first tile, where locations of the first inverter and the second inverter are associated with a second edge of the first tile opposite the first edge.

[0141] Aspect 23: The apparatus of any of aspects 21 through 22, further including: a first tile including the pillar and a second tile including a second pillar extending through the plurality of levels of the memory array, the first tile adjoining the second tile at a first edge of the first tile and a first edge of the second tile, where locations of the first inverter and the second inverter are associated with the first edge of the first tile and the first edge of the second tile.

[0142] Aspect 24: The apparatus of aspect 23, where the circuit further includes: a third inverter coupled with the signal node and a third gate line of the second tile; and a fourth inverter coupled with the third inverter and the third gate line of the second tile, where locations of the third inverter and the fourth inverter are associated with the first edge of the first tile and the first edge of the second tile.

[0143] Aspect 25: The apparatus of any of aspects 21 through 24, where to couple the bit line with the pillar, the apparatus is configured to bias the first gate line and the second gate line to a first voltage based at least in part on applying the signal to the signal node.

[0144] Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, the signal may represent a bus of signals, where the bus may have a variety of bit widths.

[0145] The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of

signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

[0146] The term “coupling” refers to condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals are capable of being communicated between components over the conductive path. When a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

[0147] The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other when the switch is open. When a controller isolates two components, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

[0148] The term “layer” or “level” used herein refers to a stratum or sheet of a geometrical structure (e.g., relative to a substrate). Each layer or level may have three dimensions (e.g., height, width, and depth) and may cover at least a portion of a surface. For example, a layer or level may be a three-dimensional structure where two dimensions are greater than a third, e.g., a thin-film. Layers or levels may include different elements, components, or materials. In some examples, one layer or level may be composed of two or more sublayers or sublevels.

[0149] As used herein, the term “electrode” may refer to an electrical conductor, and in some examples, may be employed as an electrical contact to a memory cell or other component of a memory array. An electrode may include a trace, wire, conductive line, conductive layer, or the like that provides a conductive path between elements or components of a memory array.

[0150] The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some examples, the substrate is a semiconductor wafer. In other examples, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or

epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

[0151] A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals. The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as a n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” when a voltage greater than or equal to the transistor’s threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” when a voltage less than the transistor’s threshold voltage is applied to the transistor gate.

[0152] The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

[0153] In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

[0154] The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various posi-

tions, including being distributed such that portions of functions are implemented at different physical locations.

[0155] For example, the various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

[0156] As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

[0157] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read-only memory (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

[0158] The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to

those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

1. (canceled)

2. A method, comprising:

performing an access operation on a memory cell that is coupled with a word line and a pillar extending through a plurality of levels of a memory array, the access operation comprising:

coupling the pillar with a bit line by biasing a first gate line to a first voltage and biasing a second gate line to a second voltage based at least in part on applying a signal to a signal node, wherein the first gate line is coupled with the second gate line via a first inverter and a second inverter; and

biasing the bit line to the second voltage based at least in part on coupling the pillar with the bit line.

3. The method of claim 2, wherein biasing the second gate line to the second voltage is further based at least in part on coupling the second gate line to the first gate line via the first inverter and the second inverter.

4. The method of claim 2, wherein biasing the first gate line to the first voltage is further based at least in part on applying a first signal to a first transistor coupled with the first gate line and the signal node.

5. The method of claim 4, wherein biasing the second gate line to the second voltage is further based at least in part on applying the first signal to the first transistor coupled with the first gate line.

6. The method of claim 2, further comprising:

biasing the first gate line to a third voltage based at least in part on applying a third signal to a first transistor coupled with the first gate line and the signal node.

7. The method of claim 6, further comprising:

biasing the second gate line to the third voltage based at least in part on biasing the first gate line to the third voltage, wherein the third voltage is greater than the second voltage.

8. The method of claim 7, wherein biasing the second gate line to the third voltage is further based at least in part on coupling the second gate line to the first gate line via the first inverter and the second inverter.

9. The method of claim 2, wherein biasing the second gate line to the second voltage is further based at least in part on applying a second signal to a second transistor coupled with the second gate line.

10. The method of claim 2, wherein the first inverter is coupled with the signal node and the second gate line, and wherein the second inverter is coupled with the first inverter and the second gate line.

11. A method, comprising:

performing an access operation on a memory cell that is coupled with a word line and a pillar extending through a plurality of levels of a memory array, the access operation comprising:

coupling the pillar with a bit line by biasing a first gate line to a first voltage and biasing a second gate line to a second voltage based at least in part on applying a signal to a signal node;

biasing the first gate line and the second gate line to a third voltage based at least in part on coupling the

pillar with the bit line, wherein the first gate line is coupled with the second gate line via a first inverter and a second inverter; and

biasing the bit line to a fourth voltage based at least in part on biasing the first gate line and the second gate line to the third voltage, wherein the fourth voltage is less than the third voltage.

12. The method of claim **11**, wherein biasing the second gate line to the second voltage is further based at least in part on coupling the second gate line to the first gate line via the first inverter and the second inverter.

13. The method of claim **11**, wherein biasing the first gate line to the first voltage is further based at least in part on applying a first signal to a first transistor coupled with the first gate line and the signal node.

14. The method of claim **13**, wherein biasing the second gate line to the second voltage is further based at least in part on applying the first signal to the first transistor coupled with the first gate line and coupling the second gate line to the first gate line via the first inverter and the second inverter.

15. The method of claim **11**, wherein biasing the first gate line to the third voltage is further based at least in part on applying a third signal to a first transistor coupled with the first gate line and the signal node.

16. The method of claim **11**, wherein biasing the second gate line to the third voltage is further based at least in part on biasing the first gate line to the third voltage, wherein the third voltage is greater than the second voltage.

17. The method of claim **11**, wherein biasing the second gate line to the third voltage is further based at least in part on coupling the second gate line to the first gate line via the first inverter and the second inverter.

18. The method of claim **11**, wherein biasing the second gate line to the second voltage is further based at least in part on applying a second signal to a second transistor coupled with the second gate line.

19. The method of claim **11**, wherein the first inverter is coupled with the signal node and the second gate line, and wherein the second inverter is coupled with the first inverter and the second gate line.

20. A memory system, comprising:

one or more memory devices; and

one or more controllers coupled with the one or more memory devices and configured to cause the memory system to:

perform an access operation for a memory cell that is coupled with a word line and a pillar extending through a plurality of levels of a memory array, the access operation comprising:

coupling the pillar with a bit line by biasing a first gate line to a first voltage and biasing a second gate line to a second voltage based at least in part on applying a signal to a signal node, wherein the first gate line is coupled with the second gate line via a first inverter and a second inverter; and

biasing the bit line to the second voltage based at least in part on coupling the pillar with the bit line

21. The memory system of claim **20**, wherein biasing the second gate line to the second voltage is further based at least in part on coupling the second gate line to the first gate line via the first inverter and the second inverter.

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